

How Certain is the AI?

**Neural Network Conformal Inference for Financial
Document Classification**

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Acknowledgement of Sources

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Signed Neva Fradd

Date 27/08/2026

Abstract

Neural network classifiers often produce seemingly confident outputs, even if the underlying prediction is unreliable. In this thesis, we investigate this uncertainty, and gain insight into the roots of this phenomenon by exploring the internal representations of transformer based models. We apply conformal inference as a framework to produce prediction sets of labels from a classifier model, along with a rigorous coverage guarantee. We explore the conformal inference pipeline across three increasingly complex datasets, investigating how the procedure performs when applied at different points in the model pipeline, from softmax, to logit, to the embedding layer. We hypothesise that classification uncertainty can be traced back to the embedding layer, where documents positioned near class boundaries in embedding space are systematically harder to classify than documents sitting cleanly within a class cluster. We gather notions of uncertainty on which to build conformal inference directly from the embedding space, comparing parametric, global measures against non-parametric, local ones, and further investigate whether clustering this space can improve prediction sets by separating ambiguous and clear examples. We find that local embedding-based scoring produces the most efficient and reliable results, and propose which methods carry genuine practical value to a user.

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1 Introduction

1.1 Motivation: The Cost of Uncertainty

Over the last decade, reliance on artificial intelligence to carry out day-to-day tasks has grown rapidly, particularly where automation offers businesses efficiency and cost savings. Its use has had extensive productive impact, from early-stage cancer detection [16] to fraud detection [12]. This shift has been widely accepted, both by individuals looking to avoid time spent on arduous, repetitive tasks, and by businesses looking to use employee time more productively. However, the increased deployment of AI in high-risk sectors - healthcare, credit decisioning, and financial reporting - has motivated the development of stronger regulations. The European Union’s AI Act [2] imposes obligations on AI systems in proportion with their level of risk, reflecting the principle that the widespread benefits of automation must be weighed in balance with the consequences of error.

This risk-versus-reward balance encompasses financial document processing. Bookkeeping and invoice classification are repetitive, high-volume, and expensive tasks; precisely the kind we should argue should be delegated to artificial intelligence. This is reflected in the UK government’s mandated move to electronic invoicing for all VAT-registered businesses invoices from April 1st 2029 [1]. This movement is explicitly motivated by the efficiency gains and improvements in tax compliance that digitalised invoicing enables, and will likely increase pressure to automate downstream classification of the invoices as well.

This pressure is not purely speculative, but practically motivated. Bardelli et al. [11], argue that automating invoice classification would substantially ease the work of accountants, demonstrating the task directly on Italian electronic invoices. Using standard classifiers and word embeddings, they show that account codes can be predicted automatically but that accuracy drops substantially, falling to 80% or lower, on the highest-cardinality, most ambiguous categories, such as received invoices. Their work demonstrates good classification accuracy on average, but does not address the separate question of when a model’s output should be trusted at all. This thesis addresses this question, automating a repetitive accountancy process using a neural network with domain-specific classes. We consider a 12-class invoice classification problem, in which a transformer-based model will process an electronic invoice and predict the account category of its bookkeeping entry.

The successful automation of this task is important. Misclassification of invoices can lead to downstream processing issues; incorrect tax filings, compliance failures, or incorrect accounts, all which are difficult and expensive to detect and resolve. This thesis does not simply revisit classification accuracy, as explored by Bardelli et al., but an arguably more important problem, in how we trust the stated classifications from a model.

1.2 The Technical Problem: Overconfident Classifiers

The central problem with the use of transformer-based classifiers is the well-documented overconfidence these models place in their outcomes. The final layer of a classifier produces a probability distribution across the candidate classes, reflecting the confidence the model has in which is the true class. We look for this output to be well-calibrated, in which case the model’s stated confidence aligns with its empirical accuracy, for example, a prediction with 90% confidence should be correct 90% of the time. Despite modern neural network’s predictive power and accuracy, they are systematically overconfident as a result of their training, which rewards confidence regardless of correctness and lacks signals for uncertainty. This misalignment becomes dangerous in high risk settings, and it has been argued that calibration should be focused on as much as accuracy when automating decisions [14]. Where overconfidence disguises misclassification, including in invoice classification, potential errors are not flagged, so misclassification goes unseen and causes downstream problems.

We look to correctly quantify this uncertainty and utilise it to flag potential errors in classification before continuing the pipeline.

1.3 Proposed Approach: Conformal Inference

Our proposed approach to mitigate the problems caused by transformer overconfidence is conformal inference, in which we replace the raw, likely unreliable, softmax output with a constructed prediction set. This set of candidate classes is guaranteed to contain the input’s true label with a user-specified probability, i.e. 95%. This coverage guarantee comes with no assumptions on the particular distribution of the data; only that the training and calibration data are exchangeable (Section 2.4.2). We can apply the conformal inference framework to any trained classifier without retraining, which makes this an efficient and practical uncertainty quantification strategy.

Conformal inference has been applied productively in many contexts, including transformer-based classifiers, usually applying the standard algorithm to the final softmax layer. A growing research field indicates that signals from the model before the softmax layer are potentially more effective for detecting unreliable predictions. This thesis focuses on the embedding space in depth, examining whether applying conformal inference to earlier model layers, rather than the softmax layer, improves prediction sets, and exploring whether the full local geometry of the embedding space can be formalised in a conformal inference framework in order to produce better-calibrated, more efficient prediction sets.

This hypothesis is based on the idea that the geometry of the embedding space reveals regions prone to classification errors, and, in particular, distinct sources of classification difficulty. Classification errors

can be typically attributed to two distinct sources of uncertainty: epistemic uncertainty (model uncertainty due to unfamiliar data, which can be reduced with more training) and aleatoric uncertainty (irreducible, genuine semantic ambiguity in the data itself). We consider the idea that these two types of uncertainty can be identified by patterns in the embedding space. A document that sits in a sparse, low density region of embedding space reflects an unfamiliar or out-of-distribution (OOD) input, an indication of epistemic uncertainty. On the other hand, a document sitting in a densely populated region where several classes overlap could reflect high aleatoric uncertainty, where it is genuinely unclear which class the input belongs to. We apply conformal inference directly to the embedding layer in order to exploit this nuanced uncertainty signal, which is removed by the softmax function.

Our interest in the embedding space geometry motivates our model choice, using Sentence-BERT (SBERT) rather than a standard BERT (Bidirectional Encoder Representations from Transformers) model. BERT is a powerful language model used for token-level language processing, but its embeddings are not trained to reflect semantics and input similarities through intra-embedding distances. SBERT modifies the BERT architecture, using a Siamese network structure which explicitly trains the embedding space to place semantically similar texts close together and dissimilar texts far apart (Section 2.1.1). This gives the representation space the precise geometric properties that will theoretically improve our conformal implementation, and help reliably detect epistemic and aleatoric uncertainty.

We build on this by exploring whether clustering inputs into distinct regions allow us to set localised conformal thresholds, in turn improving class-conditional coverage and producing tighter prediction sets. We explore clustering by classes following Mondrian conformal inference and begin to consider the possibility of clustering the embedding space directly into regions of high certainty (clear, well-separated areas) and high uncertainty (low-density or overlapping areas), tailoring conformal thresholds to distinct difficulty levels.

1.4 Research Questions

Through our three datasets, with the motivating application of invoice classification, we aim to explore the following questions:

1. Can conformal inference provide reliable, calibrated uncertainty estimates for overconfident neural network classifiers?
2. Does the internal geometry of a classifier’s embeddings explain and predict classification difficulty?
3. Can specific clustering methods applied to the embedding space improve conformal inference performance?

4. Which properties of a particular method indicate genuine practical value to a user relying on its predictions?

The above questions build on each other, firstly asking whether conformal inference is applicable in this setting; i.e. whether the process delivers valid, useably efficient coverage on a domain-specific classification task. We then explore the claim that conducting conformal inference on the embedding space, rather than on the softmax output, yields better-calibrated and more practically deployable uncertainty estimates, evaluated via coverage, efficiency (prediction set size), and class-conditional adaptiveness. Developing our research into the embedding space, we explore clustering algorithms, investigating how different methods of clustering embedding points affect conditional coverage of conformal inference sets. A key dependency in clustering is that the representation itself must contain coherent structure, which we investigate directly through visualisations and geometric analysis of the embedding space to ensure a clustered structure is present. Finally, rather than assuming that practical usefulness is purely determined by validity or efficiency, we look to identify which combination of output properties would indicate genuine value for a user.

1.5 Contributions

1. We apply an SBERT model to the invoice dataset and extract final layer embeddings.
2. We apply 11 different variants of conformal inference, applied at three different points in the model pipeline; embeddings, logit, and softmax and run a rigorous comparison across 8 metrics to investigate whether embedding-based conformal inference outperforms softmax-based.
3. We survey and empirically compare several clustering approaches applied directly to embedding points, evaluating their effect on the conditionality of the coverage guarantee.
4. We identify which combinations of evaluation metrics indicate genuine practical value to a user, and assess our conformal inference methods against them.

1.6 Thesis Outline

Chapter 2 reviews the literature which underpins this thesis, introducing transformer-based classifiers and the source of their overconfidence, the current landscape of uncertainty quantification methods and how they fall short before introducing conformal inference, the underpinning theory and the non-conformity scores and variants used throughout. Chapter 3 describes our datasets, the models, and the conformal inference pipeline we apply in each case. We explain the choice of evaluation metrics, used to assess coverage, efficiency, and adaptiveness. Chapter 4 explains and evaluates our results across three datasets, Fashion MNIST (used to

validate and clean the pipeline), AMuRD (used to explore pipeline behaviour on harder data), and a real dataset of invoices. Chapter 5 draws these results together, answering the questions laid out in Section 1.4, before discussing the limitations of our work and directions for future research.

2 Background and Related Work

In the Introduction, we outlined the operational risks of deploying automated document classification systems and highlighted that modern transformer-based models are prone to overconfidence and miscalibration. This chapter reviews the literature around this subject and establishes the theoretical foundations for our proposed solution. We begin by examining the architecture and training objectives of transformer-based classifiers which drive neural network overconfidence. We then survey existing uncertainty quantification methods before presenting the formal guarantees and mechanics of our specific approach; conformal inference, and we evaluate representation-space geometry to motivate our embedding-based approach.

2.1 Transformer-Based Classifiers

Before covering the formation of prediction sets applied to classifier models, we explain the models themselves. For computing tasks involving the processing of sequential data, the state-of-the-art models are transformer based [49]. For text classification specifically, fine-tuned transformer language models consistently outperform traditional methods across both single and multi-label benchmarks [25] .

The transformer architecture was introduced by Vaswani et al. [49] to overcome the limitations of recurrent neural networks (RNNs) in which sequences were processed sequentially, one token at a time. In an RNN, each token’s representation depends directly on the preceding token through a chain of hidden state updates. This sequential dependency structure poses two fundamental constraints. Firstly, since the representation for a token can only be computed once the prior token’s representation is complete, training time is directly scaled with sequence length, preventing parallelism during training. Secondly, RNNs lack the infrastructure to understand and process long-range dependencies between tokens [13]. This limitation comes from the neural network training process, in which information about a token must be propagated sequentially across the entire sequence gap. The gradients used to update model weights are calculated by repeated multiplication of the weight matrices and the activation function derivatives during backpropagation. When these values are less than one, the gradient signal shrinks exponentially, resulting in “vanishing” gradients, and as a result, the model fails to update weights for long-range dependencies. Developments such as LSTM (Long-Short Term Memory) [30] and GRUs (Gated Recurrent Units) [19] targeted the vanishing gradient problem by including “gates” between tokens to retain context, and factorisation tricks were developed to

target computational efficiency in lieu of parallelism [37]. Despite these improvements, the aforementioned limitations still remained due to the recurrent structure.

Transformers address the source of the limitations by removing the recurrence structure and replacing it with a self-attention mechanism at every model layer. Instead of processing tokens sequentially, self-attention allows the model to process all tokens simultaneously, with each token attending directly to all other tokens in the sequence. This produces a weighted representation of the sequence reflecting how relevant each token is to the one currently represented, capturing all contextual relationships between tokens despite the distances between them in the input. Therefore, long-range dependencies are no longer propagated along the sequence but captured in a single attention computation. As this process is independent of any other outcomes, it can be performed in parallel across the entire sequence, resulting in significant improvements in training time.

To process the input data without recurrence, transformers represent input data using dense vector embeddings augmented with positional encodings, which explicitly encode information about the position of the tokens into the sequence. However, because self-attention requires every token to attend to every other token, the computational and memory costs scale quadratically with the sequence length, $O(N^2)$. This means that, although long-range dependencies are better captured with self-attention, we must scale the context window to accommodate long sequences, making the models expensive and resource-intensive to run.

2.1.1 BERT and SBERT

The combination of parallelism during training and context tracking has made transformers the dominant architecture for sequence classification tasks. In particular, we use the BERT (Bidirectional Encoder Representations from Transformers) family of models [22], an encoder-only transformer architecture specifically designed to produce representations of input text for downstream tasks such as classification. The bi-directionality of BERT models gives the model the ability to assess context of a word by considering all words before and after it simultaneously, rather than just in one direction, as earlier ‘left-to-right’ language models would. This quality is developed through masked language modelling (MLM) during training, in which approximately 15% of input tokens are masked, and the model is trained to predict these tokens using the entire surrounding context. BERT also learns relationships between sentences through next-sentence prediction (NSP) during training, where a model judges whether two given sentences follow each other in some text.

These mechanisms give BERT models high-level understanding of inputs, making these encoder-only models well-suited for classification tasks in which we require the model to make a judgement (e.g. assign a label) about an input as a whole, typically assessed from a specific summary token ([CLS]).

A limitation of standard BERT is that its representations are not directly comparable to one another, meaning if we are assessing similarity between two inputs, we must pass both through the model together, as a pair. This is highly computationally expensive when many comparisons are required, for example when searching or clustering over a large collection of texts. SBERT (Sentence-BERT) [43] was developed to address this issue, producing a single, fixed-length embedding vector in representation space from an input sentence which can be compared directly to any other, without an additional pass through the network. SBERT applies a Siamese network in which two copies of the same BERT encoder model process a pair of sentences independently of one another, producing two mean-pooled vector outputs representative of the whole sentence. SBERT is trained via a contrastive objective, in which cosine similarity between the two sentence embedding vectors is computed (the angle between them), and is compared to a human’s “similarity score” between the two sentences, which comes from semantic textual similarity (STS) benchmarks. The loss between the cosine similarity and the human score is computed, and the signal is backpropagated through both the model passes (for each sentence) into the shared model weights. Through this training process, pairs of sentences labelled as semantically similar are pushed closer together in the resulting embedding space, and dissimilar pairs are pushed further apart. From using STS benchmarks, SBERT’s embedding space is organised around a human notion of semantics, and distance measures accurately reflect semantic meaning. Most literature applying conformal inference to text-based classifiers focuses on the BERT family of models, [50] identifies an improvement in the performance of prediction sets when conformal inference is applied to early and intermediate layers of BERT due to the higher level of semantic meaning encoded in these layers. This motivates our use of SBERT when we come to construct embedding-based non-conformity scores in Section 2.7.2, since SBERT’s final layer is explicitly trained for semantic meaning.

However, despite their high classification accuracy, these encoder-based models are still prone to over-confidence in their outputs. [28].

2.2 Overconfidence of Neural Networks

The overconfidence of modern neural networks, including transformer-based classifiers, is well-documented as a serious vulnerability during real-world deployment. Despite achieving significantly higher accuracy, modern architectures are, on average, notably more mis-calibrated than the shallower, more basic, models that preceded them [28], and deep networks have also been shown to produce confident results on unfamiliar inputs [41]. This pattern can be directly explained by the training methods for these models.

Models are trained through minimisation of the cross-entropy (CE) loss with gradient descent, a negative log-likelihood objective function minimised as the predicted probability of the true class approaches one (i.e.

as the model softmax output approaches the one-hot encoding of the true label). This is an asymptotic target; optimisation and a softmax output of one requires the corresponding logit value to diverge to infinity. Therefore, CE loss continues to reward further sharpening of already correct predictions forever. This is demonstrated concretely by [28] where training loss on correctly classified examples continues to decline after training accuracy plateaus, as the model can continue to reduce loss by pushing towards the one-hot outputs, despite not learning anything new. This loss function has no mechanism to reward an uncertain prediction, even if justified; as long as examples are classified correctly eventually during training, there is no differentiation between ambiguous and confidently certain input examples. The objective function does not have a signal to distinguish confidence as a reflection of true certainty from confidence resulting from continued optimisation. As a consequence, any inherent uncertainty quantification from these models is entirely arbitrary, and reflects training duration rather than a genuine measure of semantic ambiguity

Despite this, cross-entropy loss remains the default training method for classification due to its theoretical optimality in both calibration and accuracy under ideal conditions: infinite training data and a correctly specified model. Also, the gradients produced during training are well-behaved and simple to optimise; simply the difference between the predicted probability vector and the one-hot label. So CE loss’s overconfidence is not quite a flaw in the objective itself, but a consequence of applying a highly expressive model to finite data, where the model continues to reduce loss not by learning true labels but instead by overfitting to the training set and extrapolating confident predictions to unseen inputs. The architecture of the classifier model intensifies this effect. The softmax function which translates our logit values into an output “probability distribution” across classes is exponential, so small differences between logits are amplified into extreme probability ratios, resulting in more common near-certain confidence.

In addition, the expressivity of modern transformer-based models compounds this further. While this thesis focuses on SLMs (Small Language Models) in particular the SBERT model `all-MiniLM-L6-v2` with ≈ 22.7 M parameters [3], this expressivity issue is amplified in LLMs (Large Language Models) which may have hundreds of billions of parameters. Even at the SLM scale, the number of parameters gives models the flexibility to drive logits to extremes, which CE loss then continues to reward. These effects are most felt by rare or OOD (out-of-distribution) examples, where the model has less genuine basis to correctly classify but applies confident labels regardless as a result of its architecture and training [29]. Softmax scores are therefore not a true reflection of how uncertain the underlying representation is, but rather a reflection of the ranking in the logits, adapted to satisfy the training objective of the CE loss function.

In our particular context; classification of financial documents, reliable classification and well-calibrated confidence measures are practically crucial to give signal on predictions to trust, and to prevent errors from moving downstream into misrouted invoices, incorrect account entries, or compliance failures. A model giving

high confidence for ambiguous documents gives no indication for human intervention, and allows errors to move further into a process where it becomes more expensive to find and correct. We look to calibrate this confidence, not only on average across a large number of documents, but also at the level of an individual document. This motivates our review of uncertainty quantification methods in order to justify our method of choice, conformal inference, which entirely avoids the calibration problem by replacing the uncalibrated softmax with prediction sets and rigorous, distribution-free coverage guarantees.

2.3 Uncertainty Quantification Methods

Uncertainty quantification (UQ) is the field concerned with calibrating neural networks, and it’s a field of active research, especially given growing concerns around model miscalibration [14]. The UQ landscape for deep classifiers splits naturally into methods which are applied post-hoc to a model, and methods which adapt the model itself.

Bayesian methods require changing the model itself by placing a distribution over model weights rather than learning a single point estimate, approximating predictive uncertainty via techniques such as Monte Carlo dropout [24]. These approximations scale poorly to large models, and the Bayesian transformer variants typically require architectural modification incompatible with using a frozen backbone [32], reasons why these methods are a poor fit for this context. Deep ensembles train multiple independently initialised models and use their disagreement as an uncertainty signal [38]. Evidential deep learning replaces the standard softmax objective with one that directly parametrises a distribution over class probabilities [46]. Applying these methods requires retraining the model which is computationally expensive, and none offer a rigorous guaranteed uncertainty level, just an improved calibration.

Post-hoc calibration, such as temperature scaling [28], leaves the trained model untouched, rescaling the softmax output to better match true accuracy on a held-out set. This is cheap but only corrects calibration on average over a validation distribution; there is no formal guarantee of improved calibration.

The key gap we observe here in the uncertainty quantification landscape is the lack of a formal guarantee. Conformal inference avoids an attempt at calibrating a single output by instead constructing a prediction set holding a guarantee to contain the true label with a user-specified probability [51], [6] only requiring the assumption of exchangeability between the calibration and test data. Conformal inference has been combined with other UQ methods with positive results [21], however we do not explore this interaction, instead focus on evaluating conformal inference alone.

For a domain-specific, high-stakes classification task such as this one, where retraining cost and the strength of the guarantee both matter, this combination of properties is what motivates our choice of con-

formal inference as the uncertainty quantification method for the remainder of this thesis.

2.4 Conformal Inference

As stated before, conformal inference produces prediction sets for test examples with a distribution-free, user-specified coverage guarantee, under the sole assumption of data exchangeability. Applied to a classifier with K possible labels, conformal inference constructs a set-valued mapping $C(X_{test}) \subseteq \{1, \dots, K\}$, satisfying $\mathbb{P}(Y_{test} \in C(X_{test})) \geq 1 - \alpha$ for a user-specified error rate $\alpha \in (0, 1)$. Our goal is to satisfy this constraint with the smallest possible prediction sets; trivially, the guarantee is met by including every label in every prediction set, so the real design problem is efficiency, not validity. The guarantee rests on a single assumption, which we now define formally.

Definition 2.1 (Exchangeability). *A sequence of random variables $Z_1, Z_2, \dots, Z_n$ is exchangeable if their joint distribution is invariant under any permutation π of the indices:*

$$(Z_1, Z_2, \dots, Z_n) \stackrel{d}{=} (Z_{\pi(1)}, Z_{\pi(2)}, \dots, Z_{\pi(n)}).$$

Exchangeability is a strictly weaker assumption than independent and identically distributed (i.i.d.) since exchangeable variables may not be independent. In this setting, we assume that the calibration points $(X_1, Y_1), \dots, (X_n, Y_n)$, together with the test point (X_{n+1}, Y_{n+1}) , form an exchangeable sequence of $n + 1$ variables, such that the test point is simply another draw from the same underlying distribution as the calibration points. Before demonstrating how the conformal inference coverage guarantee holds on such a weak assumption, we will outline the standard split conformal inference procedure.

The three overarching types of conformal inference procedure differ between the assumptions they make about the data, and how much available data they reuse across predictions. The first developed method, full conformal inference [51] constructs a prediction set $C(x)$ for a test point x by treating each input-candidate label pair (x, y) as if it were a temporary additional data point. The model is refit (or the reference statistic (i.e. Centroid) is recalculated) on this augmented dataset, and we check if y 's non-conformity score is small enough relative to the others for the label to be included. Repeating this process for every candidate label and test point uses the available data efficiently, but the computational cost scales with the number of classes and test points, as no score or threshold can be reused. Split conformal inference [6] avoids the computational burden by dividing the data into three; a training set used to fit the model, a calibration set used to compute non-conformity scores and a conformal threshold $\hat{q}$ under the fixed model, and the held out test set. The threshold is then reused, unchanged for all test points. This does sacrifice some statistical efficiency, especially with less data since we require a calibration set to calculate $\hat{q}$, however this is our chosen

method of conformal inference (Section 3.3.1). Adaptive conformal inference [26] removes the assumption that must hold for the two methods above in which data must be a single, static, exchangeable population. This method is designed for settings in which data arrives over time, and is prone to distributional shift. Coverage is maintained by updating the effective target coverage level, responding to recently observed coverage, rather than relying on a single calibration threshold fixed in advance.

2.4.1 Conformal Inference Procedure

Conformal inference works by setting aside a held-out set of pre-labelled examples to give us an idea of how uncertainty is distributed in the dataset. Before the model sees a new input, we consider how “surprised” the model is by many examples’ true labels, demonstrating how implausible the model might find an answer, even when it is correct. We include the labels which sit below a certain level of implausibility in our output prediction set. This process can be described informally in four steps [6]:

1. Choose how to represent uncertainty in the pre-trained model. This signal may come from softmax probability, or from an earlier point such as the input’s position in the embedding space. This needs not be a calibrated probability, only an idea of how certain the model seems
2. Turn the signal into a non-conformity score $s(x, y)$ that describes how implausible label y looks for an input x , where a larger score indicates a worse fit. The most simple score is one minus the model’s predicted probability for y , so a confidently predicted label scores near zero, and an unlikely one scores close to one.
3. On a held-out calibration set of correctly labelled examples, compute how implausible the model found each examples true label (calculate $s(x_i, y_i)$, for $i = 1, \dots, n$. Sort these, and take the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score value, giving the conformal threshold $\hat{q}$, which is (just above) the $1-\alpha$ quantile of results.
4. For a new test point, build the prediction set by including every candidate label whose score falls at or below $\hat{q}$, creating a set of labels which are plausible enough to be the correct label.

The standard and simplest implementation of this procedure directly uses the classifier’s own softmax score, scoring a candidate input-label pair by $s(x, y) = 1 - \hat{\pi}(x)_y$ one minus the model’s softmax probability for the true label y . This score is called the Least Ambiguous Classifier (LAC) [45]. To build the prediction sets, we include all labels y with corresponding score below the threshold computed during calibration. When using the LAC score $s(x, y) = 1 - \hat{\pi}(x)_y$, the rule for inclusion in the prediction set is equivalent to including every label whose softmax probability satisfies $\hat{\pi}(x)_y \geq 1 - \hat{q}$.

This score is often used due to its simplicity, and how cheap it is to implement. It also often produces small average prediction sets among different non-conformity scores. However, LAC gives very uneven coverage across subgroups of inputs; poor conditional coverage. To achieve our goal of correctness across all classes or input difficulty levels, we look to implement more advanced non-conformity scores. Much current research looks to build around the (baseline) LAC score, rather than replace it directly; applying transformations to LAC’s score [35] or applying calibration procedures to the threshold and leaving the score [23] [39].

2.4.2 The Marginal Coverage Guarantee

Now we understand the procedure, we return to the question of how the coverage guarantee holds under the sole assumption of exchangeability, regardless of the score function chosen, the underlying model, or the data distribution. Let $s_i = s(X_i, Y_i)$ denote the non-conformity score of a calibration input x_i evaluated at its true label Y_i for all $i \in \{1, \dots, n\}$, and let $s_{n+1} = s(X_{n+1}, Y_{n+1})$ denote the test point’s score at its true label. Our data is assumed exchangeable, and we have a fixed score function. Since applying a fixed function to an exchangeable sequence preserves exchangeability, the scores are exchangeable random variables. Now every permutation of the sequence is equally probable, meaning the test example’s score is equally likely to sit at any of the $n + 1$ possible ranked positions. In particular, the rank of s_{n+1} among all $n + 1$ scores is uniformly distributed on $\{1, \dots, n + 1\}$. A full derivation of this proof of guarantee is given in Appendix D of [6].

By construction, $\hat{q}$ is the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest of the n calibration scores, so at least $\lceil (n+1)(1-\alpha) \rceil$ of the $n + 1$ scores fall at or below it. Combined with the uniform-rank property above, this gives directly:

$$\mathbb{P}(s_{n+1} \leq \hat{q}) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geq 1 - \alpha.$$

Recall we include labels in the prediction sets when their score $s_{n+1} \leq \hat{q}$. Following this, we have the coverage guarantee;

$$\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha.$$

Since this guarantee depends only on data exchangeability, and does not refer at all to the quality of the classifier or the non-conformity score, the validity of prediction sets is automatic. The difficulty lies in producing efficient and adaptive sets; maintaining minimal set size while still truly reflecting distinct levels of uncertainty from input to input.

2.4.3 Efficient and Adaptive Prediction Sets: APS and RAPS

In order to produce prediction sets that are functional, and give a helpful notion of uncertainty, we aim to use a non-conformity score which accurately reflects model uncertainty. LAC can be unreliable due to its focus on only a single label, ignoring the softmax distribution across other labels. This produces poor conditional coverage, over-covering easy examples and under-covering more ambiguous examples, but maintaining the correct marginal coverage guarantee across the whole test set [44]. Consider this simple example, with conformal threshold $\hat{q} = 0.95$. An ambiguous softmax output, eg. 30 labels, all with softmax $\simeq 0.03$, and all < 0.05 would give an empty set, which doesn't reflect the high uncertainty. Compare this to a confident output over 4 classes; $(0.8, 0.1, 0.07, 0.03)$ which would give a prediction set of size 3, not reflecting the confidence in the softmax. Neither of these prediction sets reflect the model's actual uncertainty due to lack of consideration on other classes. Adaptive Prediction Sets (APS) [44] gives an alternative non-conformity score which considers the probability distribution, and brings our guarantee closer to conditional coverage. For each calibration example, APS adds classes to the prediction set with decreasing softmax probability and stop when we reach the true class. The accumulated probability mass included in the prediction set once the true label has been added is the non-conformity score, where we then take the conformal threshold of all calibration examples, and use the standard rule; $\{y : s(x, y) \leq \hat{q}\}$ to form our prediction sets. Stated precisely, if $\hat{\pi}_{(1)}(x) \geq \hat{\pi}_{(2)}(x) \geq \dots$ denotes the sorted softmax probabilities for input x , and a label y occupies sorted position $o(y, \hat{\pi}(x))$,

$$s(x, y) = \sum_{i=1}^{o(y, \hat{\pi}(x))} \hat{\pi}_{(i)}(x)$$

In this score, we also include a randomisation term $U \sim \text{Uniform}(0, 1)$, which is the fraction of the probability mass of the last label added that we include in the score. Without U , the score would be the cumulative sum inclusive of the full true-class probability, creating discreteness in the scores, resulting in an over-conservative quantile threshold. By multiplying the true-class probability by U , we smooth out the discreteness, allowing us to achieve exact coverage. For this to result in a valid conformal guarantee, U must be drawn i.i.d. for each calibration and test point, and applied in the same way during calibration and testing.

For particularly ambiguous inputs, in which the tails of the softmax probability distribution are often noisy, APS prediction sets may become so large they are unhelpful. Regularised Adaptive Prediction Sets (RAPS) [5] targets this weakness by penalising classes that sit far down the sorted list by including a regularisation term to the APS score; $\lambda \cdot \max(r(y) - k_{reg}, 0)$. Here $r(y)$ denotes the rank of the true class y in the sorted probability list and the two hyperparameters determine the penalty weight per rank position (λ) and at which point the penalty is applied (k_{reg}).

This adaptation creates more meaningful and usable prediction sets on hard examples by dis-including low-probability “tail” classes. This however does include a tuning split of the data, reducing the calibration set size, which can affect results on limited data.

Another adaptation to APS is SAPS (Sorted APS) [33] which discards most probability information and focuses on the rank of the label. We discard this score as it discards information to favour simplicity, not a focus for this thesis. It is preferred for LLMs and large scale classifiers.

LAC, APS, and RAPS demonstrate that the usefulness of a prediction set is directly dependent on how much genuine information about the model’s uncertainty the chosen non-conformity score is able to capture. We follow this thread by exploring non-conformity scores which potentially capture more information than any of these three. Since softmax discards information about the input that was present in the preceding model layers, it may not be a completely faithful reflection of the model’s uncertainty (Section 1.2), motivating the idea to explore scores based on uncertainty notions that are present in the layers before, and whether these produce a more useable prediction set. We explore this in Sections 2.6 and 2.7 as well as exploring whether epistemic uncertainty, arising from limited training data or from encountering inputs unlike those it was trained on [34] is more visible in earlier layers

We look further than softmax-based scores to alternative points in the model pipeline where we can calculate non-conformity using different properties contained in the earlier layers.

2.5 Transformers Information Pipeline

Our goal is to find a non-conformity score that reflects the most information about the input – not a reflection of model training like softmax is. To explain why we look further back in the transformer layers, towards logits and embeddings, we take a closer look at the information pipeline through transformer architecture from input to classification. When an example is entered into a transformer it is converted to an embedding, a high dimensional vector representing the document. Different models are trained to carry different information through the embedding, but the key point is that nothing task-specific has been discarded. The embedding carries the signal that separates the K classes, as well as information about where a point sits relative to the rest of the training distribution, which could represent semantic similarity, document layout, or writing style. After the final embedding stage, the embeddings are mapped to a vector of K logits with the specific task of encoding how well the example fits each of the classes. Our model produces 384-dimensional embeddings as its final layer - larger BERT models often use a 768-dimensional representation space. This mapping is a linear projection learned through training; $We(x) + b = z$, where W is a $K \times 384$ dimensional weight matrix, and b its bias. Each logit is the dot product of the corresponding row in W with the embedding vector

$e(x)$. At this point we notice our first discard of information. In an embedding space of 384 dimensions, this mapping reduces dimensionality to K . Any embedding variations which exist in the null space of W , i.e. in the $384 - K$ directions, does not survive this projection, as the information encoded here is not useful for discrimination between the K classes despite possibly holding meaning about the input. The key information discarded here is the local-density signal, the directional variance of class clusters, and any within-class multimodality or off-task structure, as these details have no effect on the K -class split.

The logits are unbounded values between $(-\infty, +\infty)$ where a high positive number indicates strong preference for the corresponding class, and a low negative logit indicates low preference. These values encode class preferences but are not interpretable values, hence we pass them through the softmax function to produce a normalised scale of values, often interpreted as a probability distribution over classes. The softmax function is as follows, where z_i is the raw input logit for class i

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}, \quad i = 1, \dots, K.$$

This exponential softmax function amplifies relative distances between logits, and discards information regarding the true scale. This is a result of the shift-invariance of the softmax function, where $\sigma(z + c \cdot 1) = \sigma(z)$ for any scalar c , where 1 represents the vector made up of ones. So, two logit vectors differing by just an additive constant will map to the same softmax value. Therefore, optimised softmax scores can be produced from many logit values, demonstrating that the overall scale of the logits (an entire dimension of logit-space information) is thrown away by construction, independently of whatever training subsequently does to the surviving relative gaps.

Now we have an interpretable and well-behaved final output in the form of the softmax vector, which makes transformer-based classifiers straightforward to train, but at the cost of significant information discarded. We previously identified that a good non-conformity score is one that holds information about the input, which motivates our choice to look back further in the architecture. Recent work [7] has identified that non-conformity scores built on pre-softmax logit information produces more efficient and adaptive prediction sets than those built on softmax alone, specifically due to the logit space retaining the scale information the softmax function discards. We explore the effects of building non-conformity scores on logits and embeddings and cover recent work in this field.

2.6 Logit-Based Conformal Inference

The softmax function preserves the ranking of labels, since it is strictly monotonic, but removes the scale of differences between them. Practically, this means we can still apply rank-based non-conformity scores such

as APS and RAPS, theoretically with added sensitivity to the information not retained into the softmax layer [7]. The notion of uncertainty here comes from a reweighting of standard non-conformity scores based on how far a point is from a familiar, low-energy region of logit space. This adaptation to the non-conformity score utilises the scale information, which is removed by the softmax function, improving the adaptiveness of the resulting prediction sets.

Akhavan Attar et al’s recent work develops a framework to measure epistemic uncertainty, using Energy-Based Models in which variable configurations are associated with a scalar valued “energy”, with lower energy corresponding to higher probability. In the context of logits, a joint energy function is applied to the logit values, and it’s output is interpreted in terms of model certainty; a low free energy indicates high certainty in the model, suggesting that the input example is familiar, and the softmax values are likely a true reflection of the probability distribution across classes. On the other hand, high free energy reflects uncertainty in the model across all classes, potentially from an OOD or difficult input, indicating overconfidence in the resulting scores (Section 1.2). This free energy is used to scale the softmax-based non-conformity scores to produce prediction sets which reflect model certainty, using the following equation:

$$s_{Energy}(x, y) = s_{Softmax}(x, y) \cdot \frac{1}{\beta} \log(1 + e^{-\beta F(x)})$$

2.7 Embedding-Based Conformal Inference

Continuing along the reverse of the information pipeline, we look to the embedding space, in which inputs are represented as high-dimensional vectors. Despite losing the ranking of classes in this space, we gain information in the form of local density, embedding magnitudes, and separation between classes. We apply embedding-based conformal inference to the final layer of an SBERT model (Section 2.1.1), motivated by the high level of semantic meaning encoded, although the kind of similarity - between a string of tokens rather than a natural sentence - is not what SBERT has been trained for. We also recognise that cosine similarity, which we use throughout, can produce arbitrary results when assessing semantic similarity [47], but continue to use this as a distance measure since it is what SBERT is trained on.

2.7.1 Geometry of Embedding Spaces

We first motivate the geometric approach, considering what uncertainty means on the representation space and how we can interpret the geometry of a 384-dimensional vector space.

Definition 2.2 (Euclidean distance). *For two embeddings $u, v \in \mathbb{R}^n$,*

$$d_{Euc}(u, v) = \|u - v\|^2 = \sqrt{\sum_{k=1}^n (u_k - v_k)^2}.$$

Definition 2.3 (Cosine distance). *For two embeddings $u, v \in \mathbb{R}^n$,*

$$d_{\cos}(u, v) = 1 - \frac{u \cdot v}{\|u\| \|v\|}$$

Cosine distance discards magnitude information, only retaining the angles between embeddings, specifically relevant for SBERT family models which are trained on cosine similarity. Both distance measures are limited in high dimensional space, firstly from assuming isotropic data, treating all 384 dimensions as equally informative, equally scaled, and independent of the others. This is likely untrue in real data; variation is inconsistent across different directions. Euclidean distance also suffers under the curse of dimensionality, outlined in [15], in which as dimensionality grows larger and scale increases, all points seem to be roughly equidistant, rendering distance measures useless.

These problems motivate a third distance measure, a rescaled Euclidean distance; Mahalanobis distance [40].

Definition 2.4 (Mahalanobis distance). *For a class with mean embedding μ and covariance matrix Σ , the Mahalanobis distance of a point x to that class is*

$$D_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}.$$

So, this measure rescales class structure by its own covariance rather than assuming a spherical clustering pattern, accepting the natural class variation along some directions, while flagging deviation in directions where it is unexpected. Despite it accounting for per-class covariance, Mahalanobis can produce unstable covariance estimates, particularly with limited data.

To get an idea of embedding position, we look at three particular quantifiable geometric features. Firstly, distance to class centroid: a measure of how far an embedding is from the mean embedding of the candidate class. This describes how typical the embedding is for that specific class. Next we look at local density: how many calibration or training points are in the surrounding region, a measure of how familiar the example point is, operationalising epistemic uncertainty. Finally, the composition of the nearest points: which label do the neighbouring points hold. This reflects aleatoric uncertainty, the model-independent uncertainty which is reflected in overlapping classes [34]. Now we have a full picture of uncertainty, alongside a notion

of typicality of classes which we can develop into non-conformity scores.

These geometric features are the basis of the non-conformity scores we develop to quantify uncertainty in embedding-based conformal inference. We also explore the different distance measures as well as applying different non-conformity scores with the aim of answering the following question; does accounting for anisotropic within-class variance actually improve embedding-based conformal inference, or is a simpler isotropic distance good enough?

2.7.2 Non-Conformity Scores

Recall a non-conformity score $s(x, y)$ is a measure of how well an example x fits to a class y . The key embedding-based non-conformity scores utilise the geometric features above; a point estimate score, with and without accounting for class covariance, (distance to class centroid) and a neighbourhood-based score (k -nearest-neighbours). As in softmax and logit conformal inference, the class centroids, covariance estimate, and neighbour set used to compute scores are fitted on a reference set disjoint from the calibration set used to compute $\hat{q}$, so the score function remains fixed before the calibration and test data are observed, preserving the validity argument in Section 2.4. By also applying and comparing the effects of Euclidean and cosine distance on the above scores we ensure that task-relevant information - which may be encoded in either magnitude or direction - is not missed.

The simplest embedding-based score is centroid distance, in which we compare a test embedding $h(x)$ to a single summary of each class; the centroid μ_y which is the mean embedding of all reference points labelled y . The non-conformity score here is simply the distance between the embedding and the centroid where Euclidean and cosine distance are defined as follows;

$$s_{Euc}(x, y) = \|h(x) - \mu_y\|^2 \quad s_{cos}(x, y) = 1 - \cos(h(x), \mu_y)$$

Under the Mahalanobis score measure, the embedding $h(x)$ and the centroid μ_y is rescaled using the covariance matrix as follows;

$$\tilde{h}(x) = \Sigma^{-1/2}h(x) \quad \tilde{\mu} = \Sigma^{-1/2}\mu_y$$

before defining the Euclidean and cosine distance measures in the same way as above.

Centroid-based scores are limited by summarising an entire class to just one point. If a class is made up of several distinct sub-clusters, the centroid may lie in a low-density area, and not be a true reflection of the class.

To address this, we use a k -nearest-neighbours (k -NN) score [42], which considers the surrounding embed-

dings directly rather than summarising the whole class by a single point. Under a chosen distance measure, we observe the k nearest points to the embedding in question $h(x)$, and count the fraction labelled correctly. The non-conformity score is essentially the local disagreement with the true label;

$$s_{k-NN}(x, y) = 1 - \frac{n_y(x)}{k}$$

This structure mirrors the “ $1 - x$ ” structure of LAC (Section 2.4), just instead using a measure directly from the representation space. This non-parametric k -NN approach avoids assuming any class-conditional distribution, or any assumption about the shape of the class. Despite the theoretical motivation behind these scores, a further problem remains in conditional coverage. The marginal coverage guarantee holds, as detailed in Section 2.4.2, but our aim is for this guarantee to hold for all points individually.

2.8 Attempting Conditional Coverage

Every non-conformity score we have considered is calibrated against a single conformal threshold $\hat{q}$, computed from the entire calibration set. This global threshold guarantees marginal coverage $\mathbb{P}(Y \in C(X)) \geq 1 - \alpha$, under the assumption of exchangeability as demonstrated in Section 2.4.2, but makes no guarantees for specific inputs. Pointwise coverage, where $\mathbb{P}(Y \in C(X)|X = x) \geq 1 - \alpha$, is essentially impossible with a finite calibration set, since the output sets will always be too wide to be informative [10], so we aim to reach a weaker notion of conditional coverage, in which we have a guaranteed marginal coverage rate for groupings across data properties.

The first weaker conditional coverage we aim to reach, and the most natural for classification problems, is class-conditional coverage. Classes are not uniformly difficult so a global threshold will be too tight for difficult sets (eg. where embeddings are sparsely distributed) and too loose for easy sets (where embeddings are tightly clustered). Classwise, or Mondrian, conformal inference [51] develops a methodology to achieve this, by partitioning the calibration set by the true label, and computing separate conformal thresholds $\hat{q}_y$ for each class y from only the examples in said class. When assessing a test point’s non-conformity score for each label y , we use the corresponding $\hat{q}_y$ value to make a decision on it’s inclusion in the prediction set. This gives the guaranteed $\mathbb{P}(Y \in C(X)|Y = y) \geq 1 - \alpha$ for each class, by the rank argument applied individually to each class. However, this coverage guarantee comes with the cost of limited calibration data. Dividing the calibration set into its classes results in few calibration points to calculate thresholds, which can produce noisy and unstable quantile estimations for the $\hat{q}$ value.

This problem of not having enough data was addressed by Ding et al [23]. In order to provide more calibration data to set stable thresholds, we can group classes into clusters based on the similarity of the

distribution of their non-conformity scores across the calibration set, and then a single threshold can be computed per cluster. Well supported classes with enough calibration data to give a stable threshold are left as singleton clusters and calibrated individually as in classwise conformal inference, but classes with less points gain statistical strength from being pooled with other, similar classes. We now have coverage which holds at the cluster level and is approximately exact for each of the classes.

The research landscape beyond Ding et al’s [23] initial clustering proposal is active. Recent work aiming for conditional coverage via clustering has generalised the idea of groupings for any protected quality beyond a single class partition [27], and this extends to groups defined over both covariates and labels [9]. Discrete groupings are abandoned in favour of continuous reweighting of the calibration set for every test point [31]. A complete survey of these methods is out of scope, but we encourage future research to explore conformal inference using these approaches (Section 6).

2.9 Practical Benefits of Conformal Inference

The implementation of conformal inference to output prediction sets, and the adaptations we have spoken about with the goal of better performance and more adaptive or efficient sets, is only worth the computational cost of implementation if it indeed has a practical advantage to a user during classification tasks. We summarise the research on the utilisation of prediction sets in practical context, relevant for the general industry shift towards human-AI teams. In practical deployment, classifier outputs are checked by experts before pipeline continuation (due to machine errors) and while a single point prediction places a psychological burden on an expert as it comes with the binary decision of trust/don’t trust, a prediction set output offers a selection of options, removes the trust-calibration burden since the expert does not have to make a judgement on the systems confidence [48]; that is guaranteed through the rank argument in Section 2.4.2. This improvement, however logically sound, has been evaluated on simulated algorithms imitating “experts” applied to existing human-labelled datasets. This is developed by Babbar [8] and Cresswell [20] with human participants assessed in the study, with Babbar evaluating the improvement of conformal sets in general, and Cresswell assessing the effects of adaptive prediction sets over fixed-size prediction sets, which tests if it is the coverage guarantee alone or the adaptivity of the set size as well which assists a human, finding that more accurate decisions come from adaptive sets. This literature establishes that conformal inference sets, specifically adaptive prediction sets, have the potential to improve human decision-making. This is general literature, and is not reflective of the domain specific financial document classification we will explore further in the thesis, so we will continue to explore whether the implementation of conformal inference offers a practical advantage in our particular setting.

3 Methodology

3.1 Datasets

We want to explore how conformal inference procedures behave on real data, in particular, a dataset of invoices. Before applying any conformal procedure to the domain-specific data, we validated our full methodology on publicly available benchmark datasets with increasing similarity to the real data. Since our data is limited in size, and with unbalanced classes, validating the pipeline on clean data gives the opportunity to debug the pipeline. This involves checking for score mis-computation, determining optimal calibration splitting, and handling empty sets. Because the initial datasets have well-established behaviour in the literature, any errors can be quickly identified. Due to this, unusual results produced from the invoices can be attributed to the properties of that data, rather than implementation errors.

3.1.1 Fashion MNIST

The Fashion MNIST dataset [52] is a widely available image dataset comprising of 60,000 training images and 10,000 test images of 10 categories of 28×28 pixel, greyscale clothing items. The categories in Fashion MNIST are balanced and unrelated, and initial exploration using this dataset was purely to carry out sanity checks on our pipeline. We found that the lack of semantic overlap between classes meant we didn't have many interesting results, leading to us downsampling the images to 16×16 pixels, as detailed in Section 4.1.1.

3.1.2 AMuRD (Annotated Multilingual Receipts Dataset)

AMuRD [4] provided a substantially harder problem than that of Fashion MNIST, and the data was in our target format; text, rather than images. The original AMuRD dataset comprises 47,720 retail receipt line items across Arabic and English items, and 44 product categories. We first restrict our dataset to the purely English subset by restricting the rows to include only those with no Arabic characters. This collapses the dataset to a 16-class problem; we find that the remaining examples are generally branded, internationally labelled products. We also remove the rows with missing fields, and merge two classes with capitalisation variant; Chips & Crackers / Chips & crackers. Finally, deduplication of exact repeated item names removes a large portion of the data, removing 28% of the English subset training pool, resulting in 8,819 rows of training data and 1,268 test examples across the 16 classes, a notably high training/ test split of $\tilde{87\%}/13\%$. The per-class counts are fairly imbalanced, with training classes ranging from 276 Bakery items to 954 in the Soft Drinks & Juice category, and only 32 Cleaning Supplies ranging to 111 Soft Drinks & Juices in the held-out test set.

A key reason to validate our pipeline with AMuRD was in order to explore the embedding space of our chosen SBERT model `all-MiniLM-L6-v2`, which we expect to behave differently from the previously used CNN due to the training differences (Section 2.1.1).

3.1.3 Invoices

The final dataset we investigate is a collection of 2,169 invoices acquired from seven different clients of a UK accountancy firm. The clients belong to different business domains, and have been provided with OCR (Optical Character Recognition) technology to extract the invoice text from PDFs and images, allowing the data to be processed as text. The invoices are correctly labelled by professional accountants. The original dataset consists of 2,423 invoices across 42 categories, however we restrict our work to the categories with sufficient data; 12 categories each consisting of over 40 samples. The class sizes are very imbalanced, ranging from 45 (Stationery & Postage) to 705 (Purchases).

3.2 Models

3.2.1 Convolutional Neural Network: Fashion MNIST

We trained a lightweight convolutional neural network (CNN) to classify the images. The CNN contains two convolutional blocks, each with two successive convolutional layers with 3×3 filter blocks, a ReLU activation function, and a 2×2 Max Pooling layer, producing 32, then 64 feature maps after each block. Following the convolutional feature extractor, an embedding head projects the flattened 1,024-dimensional feature vector down to a 128-dimensional representation, with 50% dropout applied for regularisation. The final linear classifier maps this embedding to the softmax output over 10 classes. We used the Adam optimiser with a standard learning rate of 1×10^{-3} which applies adaptive learning rates, unlike SGD (stochastic gradient descent), reaching convergence in model parameters faster. The model was trained over ten epochs, with batch size 64 and CE loss as the training objective. We normalised the input images to encourage faster and more reliable training using the Fashion MNIST documented mean (0.2860) and standard deviation (0.3530).

3.2.2 SBERT Model: AMuRD and Invoices

The text from the receipts or invoices is extracted and processed as a single string of tokens, used as input into the model. We use the small frozen sentence-embedding model `all-MiniLM-L6-v2` to encode our text-based data into 384-dimensional representations. This model is from the SBERT family (Section 2.1.1), trained specifically to produce semantically meaningful representations of inputs. The model has ~ 22.7 M

parameters, making it fairly lightweight. Our embedding-based conformal procedures are applied directly onto this 384-dimensional representation space. On top of the encoder, we fit a small trainable classifier head, similar to that applied to Fashion MNIST; a $384 \rightarrow 128$ linear layer with ReLU and 30% dropout, followed by a $128 \rightarrow 16$ linear classifier. This contains 51,344 trainable parameters overall with the SBERT model itself held frozen throughout. This particular model was chosen due to its specific training objective producing semantically meaningful representations which we hypothesise will provide usable signal for the conformal procedure.

The training data used to fine-tune `all-MiniLM-L6-v2` comprises of over 1B sentence pairs, with cosine similarity assessed for every possible sentence pair. We note a limitation in our adoption of this particular model is that the training is on natural sentences, rather than a string of tokens from an invoice or receipt, which may limit how well the semantics are reflected in the embeddings.

We apply all conformal procedures post-hoc to the same fixed, pre-trained model, with the idea that every method can be compared purely focusing on the adaptations to the conformal and clustering processes.

3.3 Conformal Inference Framework

The documented procedure of conformal inference was laid out earlier in Section 2.4.1, along with an in-depth motivation of different non-conformity scores. A recap of the overall framework is as follows;

1. Define a non-conformity score $s(x, y)$ as a quantification of how well the label y describes the input x , where a lower $s(\cdot)$ means a well-suited match.
2. Hold out a calibration set of data from the test set with known labels. Compute the non-conformity score $s(x, y_{true})$ for each example x 's conformity with it's true label y_{true} .
3. Compute the $1 - \alpha$ quantile of the calibration non-conformity scores as the conformal threshold, $\hat{q}$.
4. For every new test example x , calculate the non-conformity score for every candidate label y_i for $i \in 1, \dots, K$. Build the prediction set $C(x)$ by including all candidate labels with score $s(x, y_i) \leq \hat{q}$.

3.3.1 Split Conformal Inference

We apply the split conformal procedure over the full or adaptive variants introduced in Section 2.4. Refitting the model, or recomputing reference statistics for every candidate label and test point, as necessary for full conformal inference, is too expensive and time-consuming for this context; both in this thesis, and in the real-world invoice-classification deployment we aim to build a pipeline for. The efficiency gains we require from automation require low processing costs, which are only supported by split conformal inference. We

also do not use adaptive conformal inference since this thesis assumes exchangeability in our fixed data, however this choice is revisited in Section 6 as the next step to deployment.

3.3.2 Data Splitting

The general model-training and conformal inference procedure pipeline requires a three-way data split of training, calibration, and held-out test data. The embedding-based scores we implement also require a “reference” portion of data separate to the calibration set which is used to calculate $\hat{q}$, (Section 2.7.2) on which to calculate the reference statistics needed; class centroids, the tied covariance matrix, and the neighbour set for k -NN. Instead of portioning our data further, we re-use the training data on which we fitted the classification head as the reference set for these statistics, satisfying the disjointness requirement and avoiding more division of our already-limited dataset. The split ratios are different for each of our datasets as seen in the table below. For the limited datasets we aim to maximise the absolute number of calibration examples to get the most stable possible scores.

Dataset	Train	Calibration	Test
Fashion MNIST	60,000	1,000	9,000
AMuRD	8,819	507	761
Invoices	1,298	431	440

Table 1: Data splits across the three evaluated datasets.

We still find a limitation in the size and imbalance of our final dataset especially with the smallest class containing only 45 examples. We follow the practice of repeated trials across multiple random partitions of the data and report the average of the resulting metrics to reduce the influence of unlucky partitions.

3.3.3 Non-Conformity scores and Distance Measures

Since different score functions perform well under different conditions, and it is not immediately clear which will be optimal in our particular circumstances due to a lack of systematic comparisons over non-conformity scores [17], we apply many different scores, detailed in Sections 2.4, 2.6, and 2.7.2, and compare their performance with a variety of metrics detailed in Section 3.5. We use LAC as a simple, natural baseline before applying APS and RAPS to test whether accounting for the full softmax distribution, rather than just the top ranked class, improves on this baseline, especially considering adaptivity and conditional coverage. We then look at applying logit-based scaling to these scores, to see whether the information discarded by the softmax function offers any improvement on the results. We then focus on three embedding-based non-conformity scores detailed in Section 2.7.2: a point estimate score, with and without accounting for class covariance (distance to class centroid), and a neighbourhood-based score (k -NN), and compare the effects

of Euclidean and cosine distance on each. As in softmax and logit conformal inference, the class centroids, covariance estimate, and neighbour set used to compute scores are fitted on a reference set disjoint from the calibration set used to compute $\hat{q}$, so the score function remains fixed before the calibration and test data are observed, preserving the validity argument in Section 2.4.2. We have used a single covariance matrix, estimated by pooling centred embeddings across all K classes, rather than estimating a separate covariance matrix per class in order to produce a stable estimate, at the cost of assuming that all classes share the same covariance structure. For k -NN, Euclidean and cosine distance are our primary comparison. We also check the effects of applying Mahalanobis distance with k -NN on the invoices dataset due to the large class imbalance.

3.3.4 Empty Set Handling

A known side-effect of several non-conformity scores is their tendency to output empty sets; a signal that the example does not “conform” well enough to any of the candidate labels. Due to the marginality of the coverage guarantee and the global conformal threshold, it is entirely possible that a particular example is harder than the typical uncertainty level in calibration, with no label seeming plausible under the non-conformity score and threshold, resulting in an empty prediction set. Under LAC, if the softmax probability is spread across many classes, the top ranked, most likely, class may still fail to clear the conformal threshold. This risk is also held by APS and RAPS, where the randomised tie breaking term applied to an uncertain top-ranked point can push the best candidate over the threshold. Also, in centroid based scoring, a point may be too far from any of the centroids to plausibly belong to any of them under the threshold – again resulting in an empty set. In our implementation, this affected a non-trivial fraction of test points across scores.

The standard solution in the literature, introduced in the RAPS reference code [5] forces the single top-ranked label into any empty set, resolving the immediate problem of a lack of output, and preserving the marginal coverage guarantee since we are only adding a label to the prediction set. We use this correction as a simple way to track performance, but note that this method is counter-intuitive to the genuine uncertainty that the model expresses, producing singleton sets for input examples where the model has demonstrated the least certainty about the label. This not only systematically shrinks average set size, an important metric that assesses performance of the conformal procedure but also disguises uncertainty. In deployment, a forced set containing the top ranked label is indistinguishable from a confident singleton set for any downstream process.

In practice, we recommend a different approach. Where the set would be empty, include every label. This is an honest representation of the model’s uncertainty, and hands the decision to a human reviewer

rather than automating the process.

3.4 Calibration Strategies for Conditional Coverage

The standard method of conformal inference calibrates a threshold by pooling every calibration data point regardless of the true label, providing a marginal coverage guarantee. We implement calibration strategies with the aim of reaching conditional coverage, as detailed in Section 2.8.

3.4.1 Classwise (Mondrian) Conformal Inference

We calculate a new conformal threshold for each class, using only the calibration examples of that class to calculate it. During testing, a candidate label y is only included in an example’s prediction set if the score $s(x, y)$ is less or equal to the specific threshold for that class y . The exchangeability argument that explains marginal coverage for standard conformal inference applies here, since a class is a subpopulation, so we have a coverage guarantee that holds independently for every class. However, by reducing our calibration examples, our conformal threshold is prone to high-variance, and can be particularly unreliable.

3.4.2 Clustered Conformal Inference

To mitigate the small-sample fragility noted above, we can group classes with similar score distributions, pooling calibration data, and calibrate a threshold per group. This is clustered conformal inference [23], organised so low-count classes can borrow statistical strength from larger classes with similar difficulty levels.

The process of deciding the clusters (i.e. finding the similar score distributions) is determined on data disjoint from the calibration data, otherwise the clustering step will observe the scores it calibrates against, and we have a failure of the exchangeability requirement. This means splitting calibration data into two, using the following formula

$$\hat{n} = \max(n_{min}, n_{\alpha}) \quad K = |x : n_c \geq \hat{n}| \quad \gamma = \frac{K}{75 + K} \quad M = \frac{\gamma \hat{n}}{2}$$

where n_c is the calibration count for a class c , K is the number of classes with enough examples to be clustered individually, γ the fraction of calibration data allocated to the clustering fold, and M the number of clusters. Classes falling below $\hat{n}$ are placed in a null cluster and calibrated against the standard global threshold. We find this formula is calibrated for problems with many more classes than ours, and using this directly led to all classes falling to null cluster. We implemented two direct changes to address this; first, building the clustering fold using a stratified sampling method of a fixed number of examples per class

(40 for Fashion MNIST, 20 for AMuRD), and second, setting the number of clusters directly rather than deriving this from the formula. This ensured every class was assigned to a genuine non-null cluster.

The procedure we implement is;

1. Split calibration data into a clustering and a calibration set, based on a pre-specified number of examples per class.
2. On the clustering split, summarise each class’s score distribution by a set of empirical quantiles.
3. Cluster the per-class quantile vectors into M groups using k -means clustering.
4. On the calibration fold, compute a $\hat{q}$ threshold per cluster using only the pooled scores of that cluster’s member classes. Any null cluster receives the global threshold.
5. For a test point x , include candidate label y in the prediction set if $s(x, y) \leq \hat{q}_{M(y)}$, where $M(y)$ is the cluster containing class y .

Coverage holds here at the cluster level by the same argument in Section 2.4.2.

3.5 Evaluation Metrics

Since validity is achievable trivially, we evaluate every conformal inference method against an extensive metric set, exploring three questions which we believe to encompass the qualities of a “good” prediction set.

1. Is the method valid? Is marginal coverage ≥ 0.95 ?
2. Is the method conditionally valid? Over all classes, and all difficulty levels, does coverage hold?
3. Is the method practically useful? Would the prediction sets help a reviewer?

3.5.1 Validity and Efficiency

Coverage measures the proportion of test prediction sets containing the true label, and the **coverage gap**:

$\text{CovGap} = |\text{coverage} - (1 - \alpha)| \times 100$, summarises this into a single number in percentage points.

$$\text{Coverage: } \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_i \mathbb{I}\{Y_i \in C(X_i)\}$$

Coverage either below or above the target is not necessarily an immediate failure. Undercoverage indicates a missed guarantee and requires checking this is not a result of an implementation error or an exchangeability violation. Overcoverage can arise for one of two distinct reasons.

First is the finite-sample correction we use to calculate the conformal threshold; $\hat{q} = \lceil (n+1)(1-\alpha) \rceil$ -th statistic, which is slightly overconservative, by an amount that shrinks as n , the amount of calibration data, grows [6]. Second is due to discreteness in the score distribution. APS and RAPS scores are built from a cumulative sum of probability, so we have a large jump in scores between where the true class is top ranked, and where it is ranked second or lower. If $\hat{q}$ sits in this gap then moving this threshold will provide no change in the sets until it moves to the next value, in which case coverage overshoots. This discreteness is worse when we have fewer classes. The randomised-tie breaking term U (Section 2.4.3) smooths within this structure, but doesn’t entirely remove the gap. Centroid and Mahalanobis scores also overcover since their score distributions are dominated by a small number of class-level statistics rather than varying with the input.

Average set size is our primary measure of efficiency; the mean number of labels in a test prediction set. Once coverage is achieved, we want this to be as small as possible. We also explore the distribution of set sizes since the average can disguise genuine operational usefulness; a method which produces mainly size 1 sets with a few large ones is more useful than one producing uniformly medium-sized sets. Empty sets can skew this value (Section 3.3.4), so we also report the **raw empty-set count**, the number of empty sets produced before correction.

3.5.2 Conditional Validity

However, we can achieve marginal coverage even with a particular class badly undercovered. We assess conditional coverage through two angles; by class, and across example difficulty levels (estimated through output prediction set size).

Per-class coverage is calculated separately within each class, the same way as marginal coverage is, and is targeted by our classwise and clustered calibration strategies in Section 3.4. **ClassCovGap** measures the largest deviation of a class from the target coverage level, capturing the worst-class behaviour efficiently rather than producing an analysis over all classes. **Size-stratified coverage violation** (SSCV) [6] tests conditional validity by partitioning test points based on prediction set size (a measure of example difficulty) and then coverage is computed within the partition, then SSCV reports the largest deviation from the target. Low SSCV indicates uniform coverage across sets, and has interpretable set sizes as an uncertainty signal. High SSCV reflects set sizes with inconsistent meaning. A negative value means every partition overcovers. **Worst-slice Coverage** (WSC) reports the lowest coverage found over many randomly drawn “slices” of the input space. It can detect failure patterns which are uncorrelated with classes. **FracUnderCov** (occasionally omitted) reports the fraction of classes whose coverage falls at least 10 percentage points below target, capturing how widespread a conditional failure is rather than how severe its worst instance is.

3.5.3 Practical Usefulness

Singleton hit rate is the proportion of size one sets which consist of the true label. In a practical setting these examples will theoretically avoid extra checks, so this number tracks the accuracy of the automation. The **adaptiveness gap** is the difference between average set size on incorrectly and correctly classified test examples. We look for a large gap, representative of the score responding to genuine per-example difficulty. This metric gives extra insight into the average set size; LAC produces the smallest set sizes on Fashion MNIST but the least adaptive.

3.5.4 Repeated Trials

We compute these metrics over many different calibration/test splits of the data, since the marginal guarantee only holds in expectation, and since our datasets are small (AMuRD and invoices) we want reporting to demonstrate properties of the underlying method rather than the sampling variability, applying 200 trials for softmax scores, 150 for Centroid and Mahalanobis, and 100 for k -NN. We assess coverage over these trials, and compare with the theoretical Beta distribution which coverage levels across splits should follow [6].

4 Results

4.1 Fashion MNIST

The Fashion MNIST dataset, introduced in Section 3.1, is a well-behaved, widely available image dataset to which we apply a simple 2 layer CNN (Section 3.2), so the results should be explainable due to well-documented properties of the data. We explore our research questions through the following results, first assessing whether conformal inference produces informative and useful prediction sets, then investigate the proposed improvements of applying embedding-based conformal inference, and finally whether we can further improve the quality of prediction sets by clustering on the embedding space.

4.1.1 Downsampling Data

After applying the CNN and a LAC pipeline to the dataset, we found the output gave 226 empty sets. This was not, as initially expected, a result of OOD inputs, but rather a result of a high conformal threshold. The CNN on such a simple dataset had achieved high accuracy on its single-output predictions, and with high confidence. Then, when applying LAC, many of the top-rated labels were true, and had corresponding softmax probability ≥ 0.95 , resulting in a conformal threshold of $\hat{q} = 0.0339$. When applied to the test dataset, any example with all softmax probabilities $\leq 0.9661 = 1 - \hat{q}$ resulted in an empty set. We chose to

downsample the Fashion MNIST images from 28×28 pixels to 16×16 pixels, creating more ambiguity, and resulting in a final model test accuracy of 0.9095.

Table 2: Master comparison of nonconformity scores on Fashion-MNIST.

Method	Coverage	AvgSize	CovGap	ClassCovGap	SSCV	WSC	SingletonHit	RawEmpty
APS	97.21%	1.276	2.21	2.70	2.39	92.22%	97.39%	129
APS+Energy	96.79%	1.279	1.79	2.55	2.19	89.89%	97.19%	179
Centroid (cosine)	95.01%	1.247	0.01	3.84	5.00	81.56%	94.31%	164
Centroid (raw Euclidean)	97.87%	2.211	2.87	3.54	3.71	89.22%	98.15%	194
k-NN (cosine)	95.80%	1.178	0.80	3.35	5.00	86.56%	96.06%	0
k-NN (raw Euclidean)	96.70%	1.159	1.70	2.71	4.09	89.78%	97.37%	0
LAC	95.50%	1.144	0.50	3.29	5.00	86.67%	95.94%	0
Mahalanobis (cosine/normalized)	97.24%	2.510	2.24	2.39	4.88	86.44%	95.45%	299
Mahalanobis (raw)	98.13%	1.966	3.13	3.26	4.05	91.33%	98.15%	232
RAPS	97.03%	1.258	2.03	2.57	5.09	91.33%	97.44%	125
RAPS+Energy	96.66%	1.259	1.66	2.48	5.00	89.56%	97.19%	179

Target coverage: 95%.

Lower is better for AvgSize, CovGap, ClassCovGap, SSCV, RawEmpty. Higher is better for WSC, SingletonHit.

WSC computed via 500 random projections of the shared 384-dimensional embedding space, $\delta = 0.1$.

4.1.2 Softmax-Based Conformal Inference

To begin with, we applied the three standard softmax-based non-conformity scores, LAC, APS, and RAPS, with an $\alpha = 0.05$ target (95% coverage). For empty raw prediction sets, we have force-included the top ranked class, a standard method (Section 3.3.4) used throughout. RAPS hyperparameters have been fixed at their default values of $\lambda = 0.2$ and $k_{reg} = 3$ since they transfer reasonably well across datasets [5].

The results are summarised in Table 2. LAC produces the smallest average prediction sets, with 86.7% of inputs receiving singleton sets but this comes at the cost of adaptiveness; the 0.590 adaptive gap represents the gap between average set size on correctly and incorrectly classified examples, indicating that LAC’s set size is not as sensitive to per-example difficulty in comparison to APS and RAPS. This is due to the LAC discarding the distribution of probability across labels. APS produces slightly larger prediction sets on average, but with a higher adaptiveness gap (0.945), meaning that this score genuinely responds to per-example difficulty, instead of applying a near-uniform threshold.

The overcoverage from APS and RAPS (97.21% and 97.03%) is attributed to the randomised tie-breaking implementation and the empty set correction (Sections 2.4.3 3.3.4). We expect a larger effect from the RAPS penalty on problems with greater genuine class ambiguity, and test this directly in the AMuRD experiments in Section 4.2.2.

4.1.3 Logit-Based Conformal Inference: Energy Scaling

Motivated in Section 2.5, we use information encoded in the logit layer to explore whether they carry non-conformity signal which the softmax function discards, applying energy scaling from logits to the APS and RAPS scores. The tuned β hyperparameter tunes how much energy scaling to apply, selecting the value which minimises average set size while maintaining target coverage.

Across the tested β values, average set size was monotonically non-decreasing with no interior minimum. The tuning procedure therefore selected the smallest available $\beta = 0.01$, which was the value closest to applying no energy scaling at all, with resulting energy-scaled scores being statistically indistinguishable from their unscaled counterparts as seen in Table 2. We suggest logit magnitude carries little usable signal beyond what is available from the rank based softmax scores. This is consistent with the literature [7], where reported gains from energy scaling are most often reported from larger architectures; not in a 2-layer CNN.

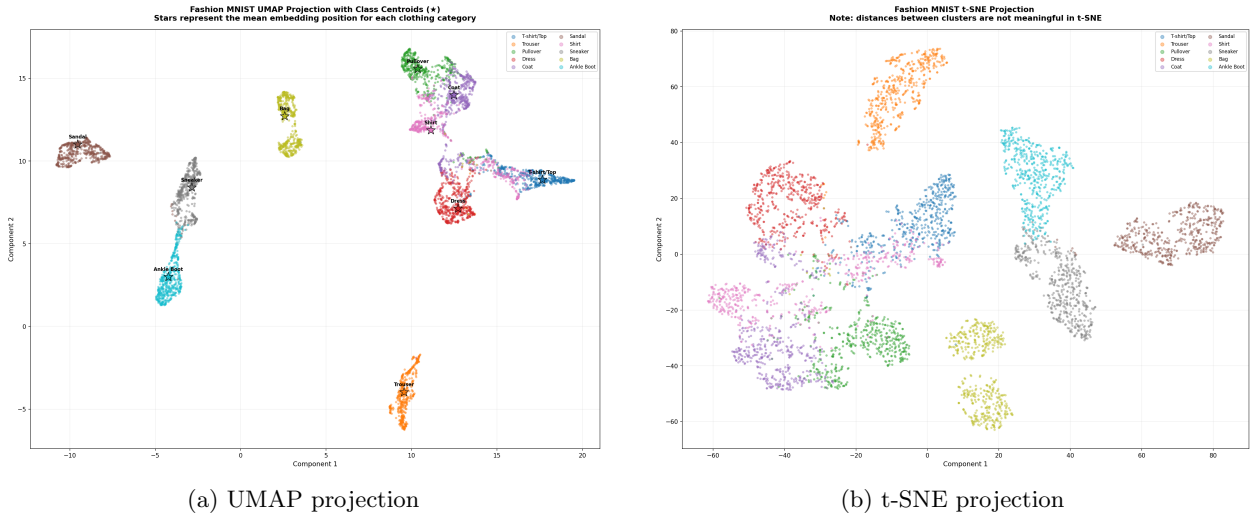


Figure 1: 2D projections of the Fashion MNIST 128-dim training embeddings, coloured by class, comparing UMAP with t-SNE as two dimensionality-reduction techniques. The consistent structure indicates that the observed class relationships are genuine, and not a result of either projection method. Mean class embeddings (Centroids) are marked with stars in UMAP. Note that distances between clusters are not meaningful in t-SNE.

4.1.4 Embedding-Based Conformal Inference

We evaluate Centroid distance, Mahalanobis distance, and a k -nearest-neighbours (k-NN) score under two distance measures (Euclidean and cosine) applied to the final layer 128-dimensional embedding. The distance measure proved important for the Centroid-based scores, in which Euclidean distance produced larger prediction sets than the cosine variant which discards all magnitude information (average size 2.211 vs 1.247).

This also comes with a more conservative coverage (97.87% vs 95.01%). We can infer that the spread of Euclidean distances (representative of angles between embeddings) is more dispersed than the cosine, meaning the Euclidean $\hat{q}$ will be larger, therefore admitting more candidate labels to the prediction set. This is likely due to embedding magnitudes carrying noise unrelated to which class they belong to. This result is reversed for Mahalanobis distance, providing larger sets under cosine distance, showing that the best-performing distance measure is not necessarily consistent. Distance measures have little effect on k -NN scoring, consistent with its non-parametric construction.

We examine the adaptiveness of these methods, comparing the prediction sets for correctly vs incorrectly classified test examples. For Centroid and Mahalanobis distance (both evaluated on nearest-centroid agreement we have $n=7,881$ correctly, $n=1,119$ incorrectly matched), the raw-Euclidean scoring produces a large adaptiveness gap of 1.864 for Centroid and 1.773 for Mahalanobis, contrasting with cosine distance (0.329 and 0.289 respectively). The k -NN measure (evaluated on 1-nearest-neighbour agreement, giving $n = 8,079$ correctly, $n = 921$ incorrectly matched) shows a comparatively modest gap (0.672 raw, 0.733 cosine).

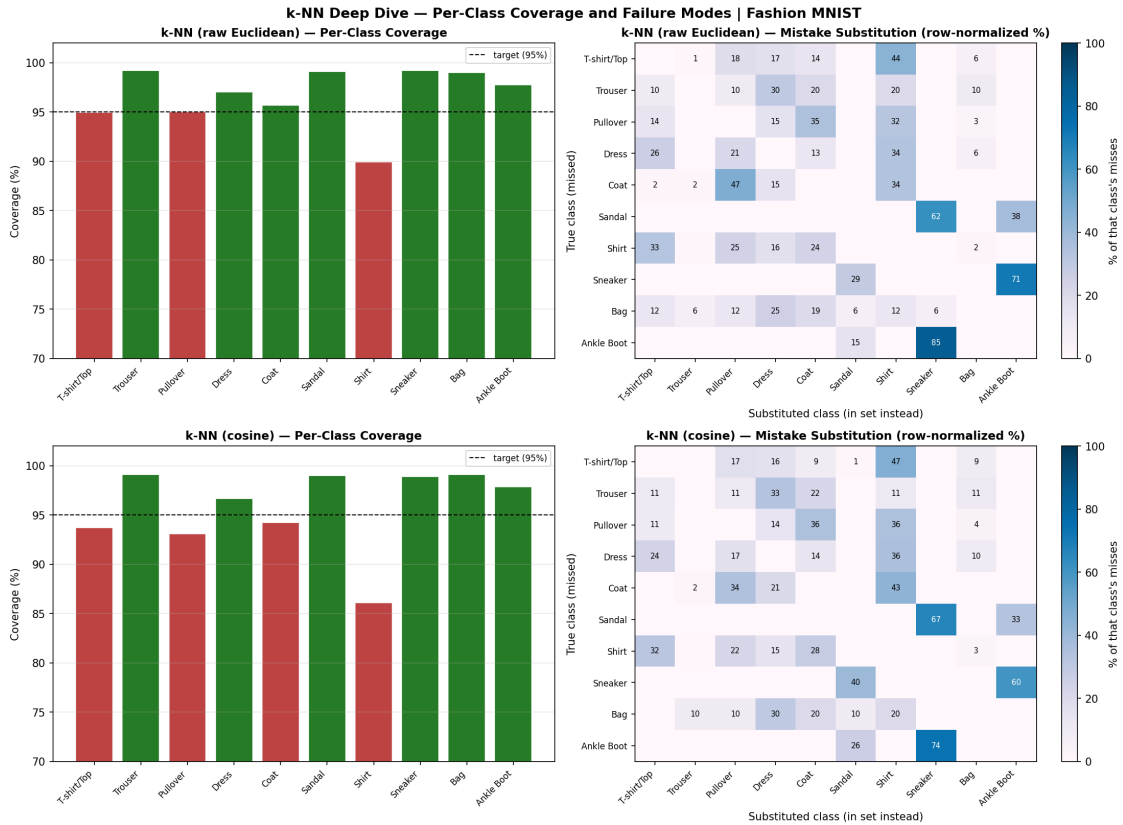


Figure 2: k -NN per-class coverage (left) and mistake substitution heatmaps (right). Red bars fall under the 95% target, Shirt substantially undercovered in both distance modes; its missed examples are replaced mainly with T-shirt/Top, Pullover, and Coat.

A key result overall is that distance measure choice matters for parametric scoring methods, but not for k -NN, a non-parametric method, since the global properties of direction and magnitude have little effect on local neighbourhood analysis (see consistent k -NN pair in Figure 5). Figure 5 also demonstrates that no single method dominates across both measures of conditional coverage (ClassCovGap and WSC). This validates tracking conditional coverage through multiple angles as they clearly show different underlying properties of the method.

4.1.5 Classwise and Clustered Calibration

So far we have demonstrated our methods reaching the marginal coverage guarantee of 95%, but this single coverage level says nothing about whether any specific class is being systematically over or undercovered. We aim to reach conditional coverage across classes, while outputting sets which are a true reflection of uncertainty in a particular class. We use the k -NN score to explore per-class coverage further.

From Figure 1a and 1b, we hypothesise that the clean-cut classes (Sandal, Bag, Trouser) should produce mainly singleton prediction sets only containing the true set, whereas examples belonging to the classes in the cluster should produce larger prediction sets as a reflection of genuine ambiguity. We look closer at these classes in Figure 3, demonstrating true ambiguity between the classes, or aleatoric uncertainty.

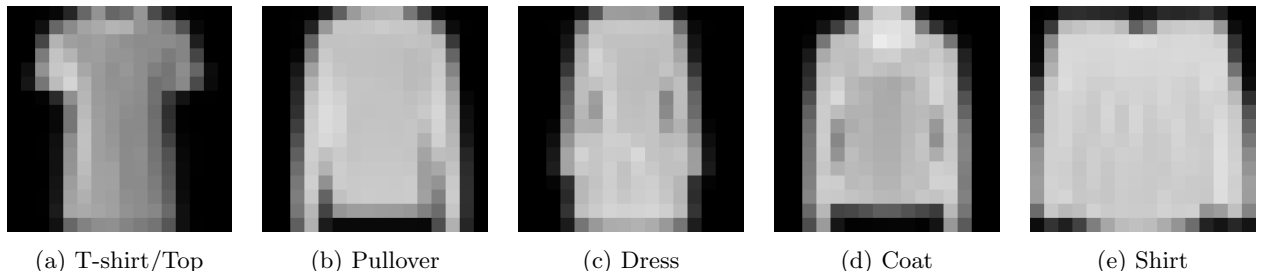


Figure 3: One example each of the five confusable Fashion-MNIST classes, illustrating the genuine visual ambiguity between them at 16×16 resolution.

We implement Mondrian and Ding et al.s calibration strategies, allocating $n_{clust} = 40$ examples per class to the clustering fold and selecting $M = 4$ clusters (Section 3.4), applied to the k -NN score with the goal of achieving conditional coverage over classes, and assess how the score behaves across the changes in difficulty levels.

Under the standard calibration method, in which a single global threshold, in this case k -NN is used for all test points, the well-separated classes (seen in Figure 1a) overachieved coverage, whereas the Shirt class was substantially undercovered (88.14% raw-Euclidean, 88.48% cosine), its difficulty reflected in the class spread in embedding space, tangled with other classes. Figure 2 shows this variability in coverage alongside a

mistake substitution heatmap, showing embedding confusion between classes. Despite undercoverage, Shirt maintains the low average set size within the class (1.531, and 1.380), a misleading result. We aim to keep a consistent coverage level across classes while producing adaptive sets reflective of the input difficulty, and attempt to achieve that now with classwise conformal inference (which sets a $\hat{q}$ per class).

The results comparing the standard procedure (using a global uncertainty threshold) against classwise and clustered are laid out in Table 3. Under raw-Euclidean scoring, classwise calibration produced a genuine, cost-proportionate correction: Shirt’s coverage rose to 96.31%, and no other class fell below target (worst class Dress, 96.04%), at a marginal 8.5% increase in average set size (1.159 to 1.257). Clustered calibration improved marginal and worst-class figures further (coverage 97.34%, worst class Bag at 96.35%) but, notably, at a worse ClassCovGap than classwise (2.34 vs. 1.93) – a reminder that worst-case coverage and class-conditional variance can move in opposite directions, and that no single summary metric captures conditional fairness completely.

Table 3: Standard, classwise, and clustered calibration compared for the k -NN score, both distance modes.

Mode	Strategy	Coverage	AvgSize	ClassCovGap	Worst class	Worst cov.
raw Euclidean	Standard	96.08%	1.159	2.69	Shirt	88.14%
raw Euclidean	Classwise	96.93%	1.257	1.93	Dress	96.04%
raw Euclidean	Clustered	97.34%	1.293	2.34	Bag	96.35%
cosine	Standard	96.21%	1.178	3.00	Shirt	88.48%
cosine	Classwise	98.30%	4.615	3.30	Ankle Boot	95.41%
cosine	Clustered	98.32%	5.491	3.40	Ankle Boot	94.62%

Under the cosine distance measure, the same adaptations produce a seemingly positive effect on the marginal statistics (classwise coverage 98.30%, clustered 98.32%) that does not reflect genuine recalibration. Observing the per-class threshold diagnostics in Figure 4 reveals why. Four classes, T-shirt/Top, Pullover, Shirt, Coat, receive a classwise threshold of exactly $\hat{q} = 1.0$, which is the k -NN mathematical ceiling. This value arises directly from the score’s construction: with $k = 5$ under cosine scoring, a calibration point whose 5 nearest neighbours in embedding space contain none of its own true class produces a score of exactly 1.0; once enough of a class’s calibration examples fall into this genuinely ambiguous region of representation space, the 95th-percentile threshold for that class saturates at the ceiling. Intuitively, the score is reporting that, for these four classes, a substantial share of calibration examples have no same-class neighbours among the 5 nearest, i.e. the learned embedding has not found a locally coherent region for these classes. This means that the resulting saturated conformal threshold is a reflection of true uncertainty rather than a defect in the calibration procedure.

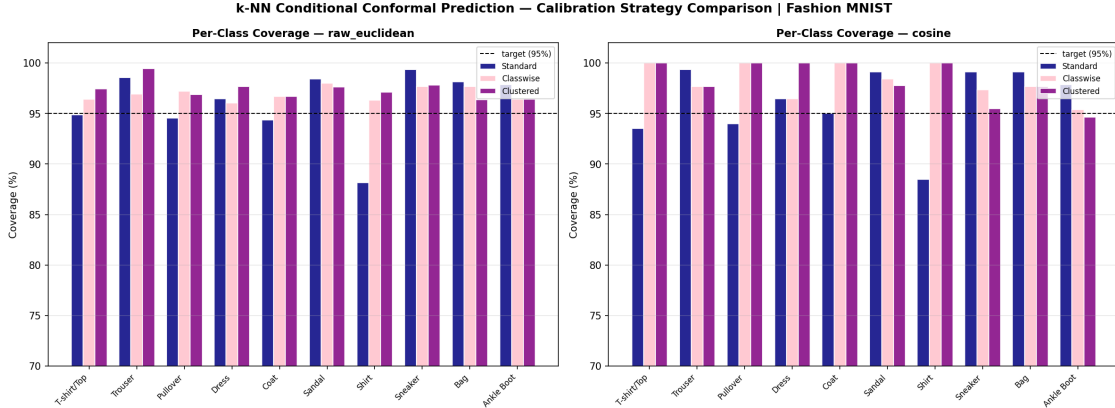


Figure 4: Per-class coverage under standard, classwise, and clustered calibration for the k -NN score, under Euclidean (left) and cosine (right). Classwise calibration brings all classes above target coverage, but increasing average set size by 8.5% in Euclidean and saturating four classes’ thresholds at 1.0 in cosine -i.e. admitting those classes into every prediction set.

A saturated threshold of 1.0 admits every candidate class unconditionally: because the score is bounded above by 1.0 for every possible input, the inclusion test $s(x, y) \leq 1.0$ is satisfied trivially, regardless of x . Each of the four saturated classes is therefore force-included in every test example’s prediction set, irrespective of the input or any supporting evidence, purely to satisfy that class’s own conditional coverage requirement – which is precisely why average set size rises to 4.615 (classwise) and 5.491 (clustered), more than triple the corresponding raw-Euclidean figures. The relative deterioration from classwise to clustered can be attributed to the clustering together of already saturated classes; Coat and Shirt. The pooled threshold must then accommodate the harder of the two classes’ scores.

This result is key to our exploration of clustering methods in the conformal inference landscape. We know a conformal procedure producing large, honest prediction sets for classes with genuine representational ambiguity is functioning correctly: it is accurately reporting the model’s inability to discriminate between these classes, not malfunctioning. Therefore, improving on this specific result requires improving the underlying representation or model architecture with a metric-learning objective that explicitly separates these classes, over a different calibration procedure. However, we look to formulate a better conformal procedure, which can distinguish the “messy” part of the embedding space, and treat this differently to the well-separated classes prior to calibration, rather than post-hoc.

We find that the clusters found are inconsistent across distance measures, only partially matching the five-class tangle identified (raw: {T-shirt/Top, Pullover, Dress, Coat} grouped with Shirt isolated; cosine: {T-shirt/Top, Pullover, Dress} grouped with {Coat, Shirt} separately). This indicates that the clustering outcome is sensitive to the score distribution it is computed from.

Also, when choosing k -NN’s neighbourhood to select the number of neighbours considered (k selected

to minimise set size while reaching coverage), the selected k (100) was the largest value in the tested grid. When we tested further, the mechanism chose even larger k values, likely because by only taking into account the close neighbours, we are capturing more of the uncertainty in the 5-class mess, which in turn increases average set size. This is another reason why we cannot simplify results to “small average set size = good” as there are more factors at play. We keep $k=100$ to observe how classwise conformal inference behaves here.

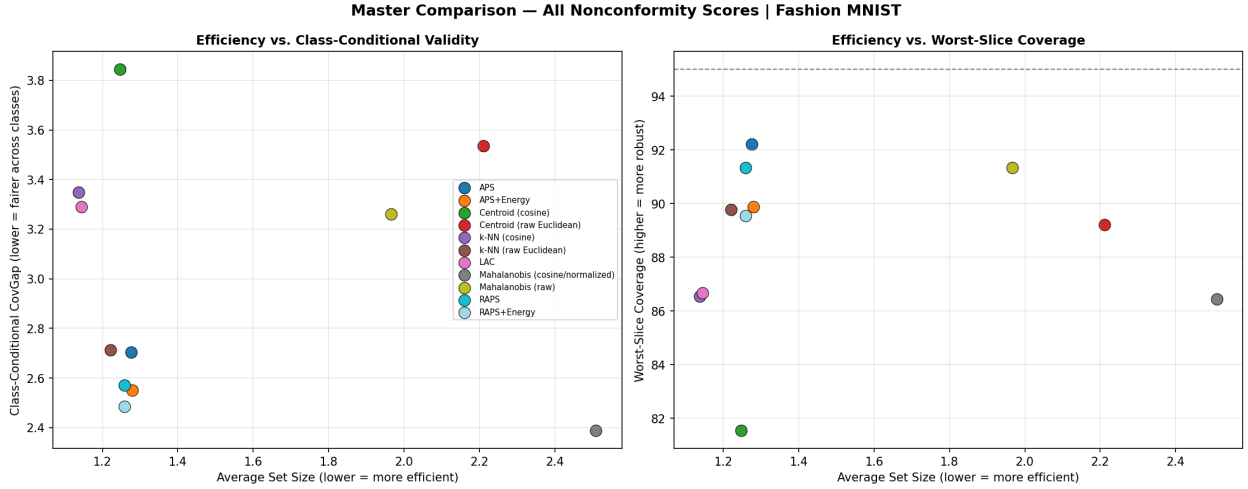


Figure 5: Master comparison of 11 scores tested on Fashion MNIST, plotting average prediction set size against two independent measures of conditional validity; class-conditional coverage (left, shows fairness across classes) and worst-slice coverage (right, tracks performance on arbitrary sub-populations).

4.1.6 Fashion MNIST Summary

Marginal coverage was achievable across all non-conformity scores with a 95% target. The informative differences lay in efficiency, set-adaptiveness, and class-conditional behaviour. We found embedding magnitude carried meaningful signal for scores built on global per-class summary statistics (Centroid, Mahalanobis) but negligible signal for the fully local k -NN construction, a distinction that recurred consistently across both efficiency and adaptiveness metrics. Classwise and clustered calibration corrected the per-class unfairness across coverage levels, but only genuinely where the score is granular enough to support real discrimination between classes. For example, where it did not (cosine-mode k -NN at $k = 5$), an apparently successful correction concealed a complete loss of information for the affected classes. Having established this pipeline, its failure modes, and the diagnostic tools needed to distinguish a genuine correction from a superficial one, we look to explore a more complex dataset with a higher similarity level to our target dataset; AMuRD, where we test which of our findings are properties of Fashion MNIST specifically, and which generalise.

4.2 AMuRD (Annotated Multilingual Receipts Dataset)

We have established the conformal inference pipeline and gained understanding of the outputs from the results on Fashion MNIST, we now apply the same pipeline to AMuRD. This is a substantially harder problem, in which the categories are not well-separated, as demonstrated in the 2D projections (Figures 6a, 6b). There are 16 text-based product categories in which each receipt item belongs to one class, built on frozen sentence embeddings from `all-MiniLM-L6-v2`.

In this section we continue to explore our research questions; first; does our tested conformal inference pipeline continue to produce reliable, useable prediction sets on this more challenging dataset, and secondly, does the embedding space contain non-conformity signal that genuinely improves prediction sets, or does the value of embedding-based scoring we found on Fashion MNIST fail to transfer to this context. We find the results from the embedding layer are more data-dependent than previously thought, and we explore these ideas further in Section 4.2.4, before conducting an exploratory clustering procedure in Section 4.2.6.

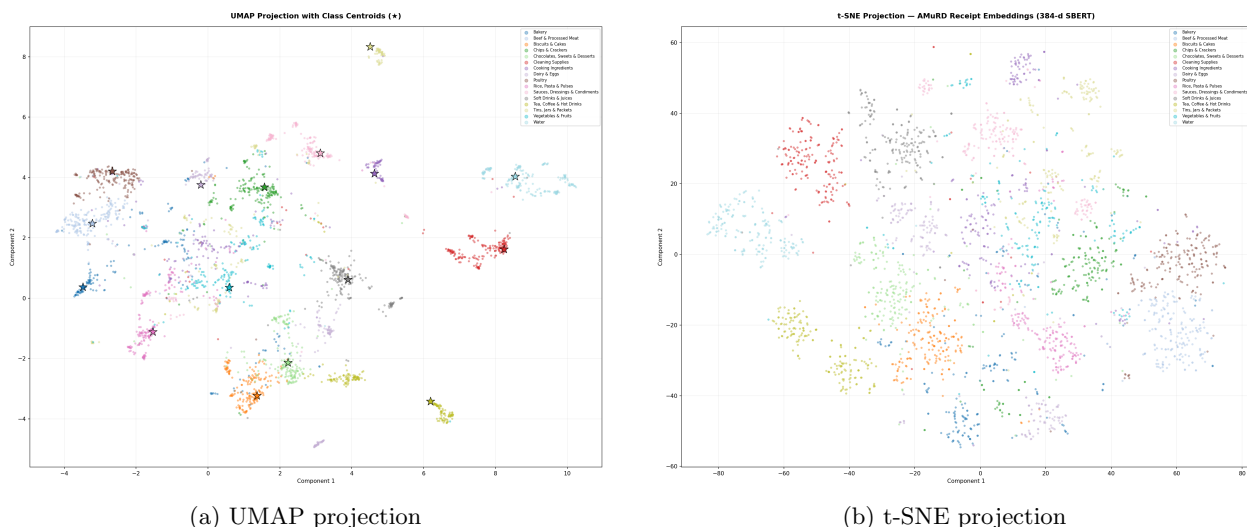


Figure 6: 2D projections of the AMuRD 384-dim training embeddings taken from the SBERT output, coloured by class, comparing UMAP with t-SNE as two dimensionality-reduction techniques. The consistent structure indicates that the observed class relationships are genuine, and not a result of either projection method. Mean class embeddings (Centroids) are marked with stars in UMAP. Note that distances between clusters are not meaningful in t-SNE.

4.2.1 Dataset and Model

The AMuRD data was introduced in Section 3.1 along with our implemented pre-processing, in which we took the English subset and cleaned the data, leaving us with 8,819 training examples and 1,268 test examples, all in the form of receipt line items across 16 categories. This dataset is significantly more imbalanced than Fashion MNIST as demonstrated in Figure 7. Test set sizes range from 32 (Cleaning Supplies) to 111 (Soft

Drinks & Juices), which does not provide much calibration data for the conformal procedure.

Each receipt item’s text is encoded using a small frozen sentence-embedding model, `all-MiniLM-L6-v2`, from the SBERT family with 384-dimensional output. On top, a small trainable classifier head was fitted, similar to the classifier applied to Fashion MNIST: a $384 \rightarrow 128$ linear layer with ReLU and 30% dropout, followed by a $128 \rightarrow 16$ linear classifier. This contains 51,344 trainable parameters overall with the SBERT model itself held frozen throughout.

We train the classifier model over 15 epochs, reaching a final test accuracy of 90.54%. This is a very close final accuracy to the Fashion MNIST model; we have aimed to tune our models to a slightly imperfect accuracy level, avoiding the high confidence which affected our conformal pipeline we saw on the original 28×28 Fashion MNIST images, as well as avoiding a weak model which would make all our considered non-conformity scores equally uninformative.

An additional challenge here is AMuRD’s much smaller overall data volume, of which we explore the practical implications of this limitation particularly in Section 4.2.5. For $n_{cal} = 507$ and $\alpha = 0.05$, we have a theoretical target for marginal coverage with mean = 95.079 and std $\approx 1.239\%$, which is our reference throughout the following non-conformity scores as the reference for the actual coverage distributions.

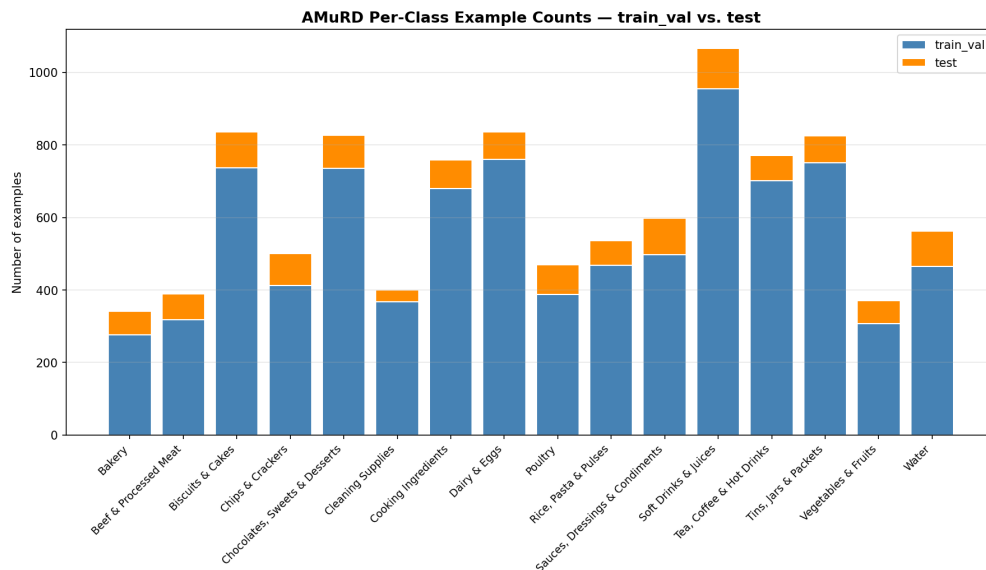


Figure 7: Per-class example counts for the AMuRD English subset after pre-processing, split into training and test data. The inconsistency in class sizes provides data imbalance which leads to classwise calibration instability in Section 4.2.5

4.2.2 Softmax-Based Conformal Inference

We explore the softmax-based non-conformity scores applied to the output of the classifier head, tracking metrics over 200 independent calibration/test splits. Coverage is met across all methods, and overcoverage in APS and RAPS comes from an inflated $\hat{q}$ (single split: 0.9385, 0.9663 vs LACs 0.8895), reflecting the discreteness in the score distribution, compounded by small class counts (Section 3.5), and leading to a slightly higher average set size (Table 4) in order to meet the coverage level, set by $\hat{q}$. We complete an extra check on coverage by producing coverage histograms, compared against a theoretical Beta curve [6] in Figure 8, where we see that LAC matches well. APS and RAPS are skewed due to small samples, as expected, but the coverage also has a narrower spread, which can also be explained by the discreteness in the score distribution forcing the scores to a particular range.

Table 4: Multi-trial coverage and stability metrics, softmax-based scores (mean $\pm$ 1 std across 200 trials; theoretical coverage target: 95.079% $\pm$ 1.239pp).

Method	$\hat{q}$: single	Coverage	AvgSize	ClassCovGap	SSCV
LAC	0.8895	95.2% $\pm$ 1.2pp	1.221 $\pm$ 0.049	3.44pp $\pm$ 0.70pp	12.24pp $\pm$ 8.85pp
APS	0.9385	98.3% $\pm$ 0.5pp	1.858 $\pm$ 0.111	3.71pp $\pm$ 0.34pp	4.17pp $\pm$ 0.50pp
RAPS	0.9663	97.6% $\pm$ 0.6pp	1.689 $\pm$ 0.086	3.10pp $\pm$ 0.34pp	18.95pp $\pm$ 7.52pp

We see a largely increased SSCV (Size-stratified coverage violation) from APS to RAPS (18.95 percentage points, maximum 40.46 over 200 trials, relative to APS 4.17). This means that the adaptivity of RAPS is not as trustworthy as APS; the coverage guarantee is not consistent across the prediction set sizes (Section 3.5). This finding is the opposite of the pattern we observed on Fashion MNIST, in which RAPS outperformed on this metric, demonstrating the inconsistencies of score methods across datasets.

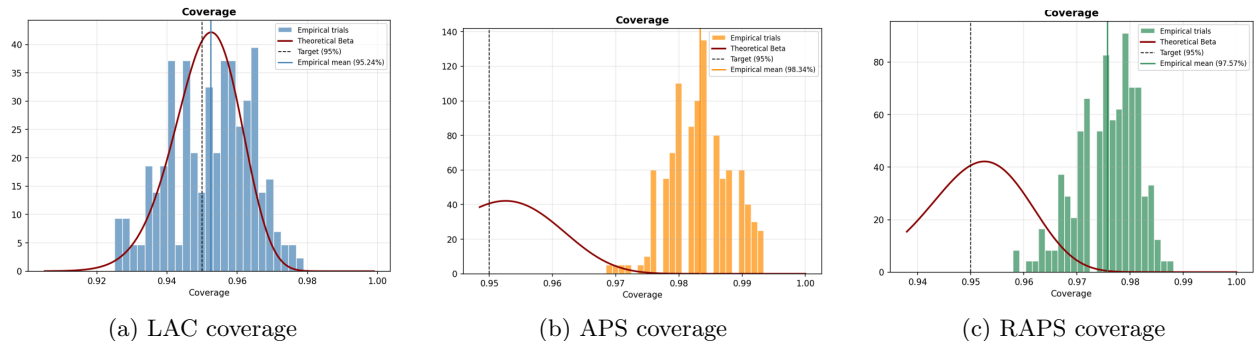


Figure 8: 200-trial coverage histograms for softmax-based scores applied to AMuRD, compared with the theoretical repeated-trial Beta distribution. These histograms are consistent with the theory, the observed skew is from discreteness in the data.

Per-Class Behaviour: RAPS Deep Dive We explore a case-study of per-class behaviour via the RAPS non-conformity score. The coverage patterns observed here are consistent with the representation space structure seen in the 2D UMAP and t-SNE projections of AMuRD’s 384-dimensional embeddings (Figure 6). The section of tangled embeddings is significantly more complex than Fashion MNIST’s single five-class tangle, and the mistake pattern across RAPS prediction sets reflects this.

A key takeaway here is that high coverage is not necessarily useful; 100% coverage on a class implies prediction sets are including said class where it is unlikely to reach that certainty.

Table 5: Per-class coverage, RAPS, canonical split ($N = 761$ test examples, 14 raw empty sets).

Class	N	Coverage	Gap	AvgSize
Bakery	36	97.22%	+2.22	2.583
Beef & Processed Meat	39	92.31%	−2.69	1.744
Biscuits & Cakes	60	100.00%	+5.00	1.750
Chips & Crackers	60	98.33%	+3.33	1.917
Chocolates, Sweets & Desserts	51	98.04%	+3.04	1.686
Cleaning Supplies	17	100.00%	+5.00	1.353
Cooking Ingredients	43	97.67%	+2.67	1.907
Dairy & Eggs	42	100.00%	+5.00	1.619
Poultry	51	100.00%	+5.00	1.686
Rice, Pasta & Pulses	35	94.29%	−0.71	1.686
Sauces, Dressings & Condiments	57	98.25%	+3.25	1.719
Soft Drinks & Juices	73	100.00%	+5.00	1.274
Tea, Coffee & Hot Drinks	43	97.67%	+2.67	1.488
Tins, Jars & Packets	52	98.08%	+3.08	2.154
Vegetables & Fruits	37	94.59%	−0.41	2.730
Water	65	100.00%	+5.00	1.200

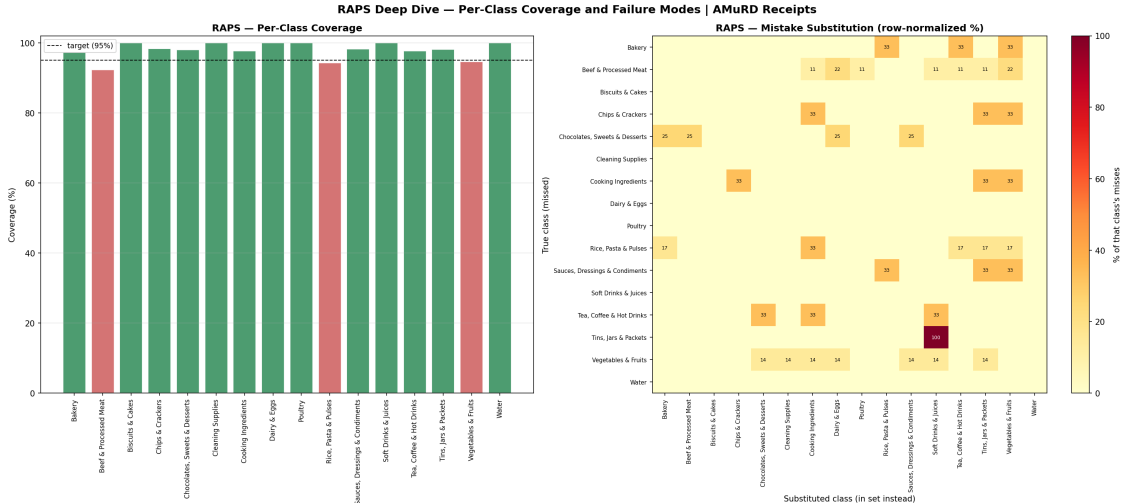


Figure 9: RAPS per-class coverage (left) and a mistake-substitution heatmap (right) on AMuRD. Three categories; Beef & Processed Meat, Rice, Pasta & Pulses, and Vegetables & Fruits fall below target, with the vast majority of substitutions plausibly semantically similar.

4.2.3 Logit-based Conformal Inference: Energy Scaling

Using the same methodology as before, we tuned β (which controls how much weight to give the logit signal) to optimise set size under valid coverage. As in Fashion MNIST, the β value selected was the smallest available (0.001), producing energy-scaled scores which were statistically indistinguishable from their unscaled versions. This strengthens our conclusion that logit-magnitude carries little relevant non-conformity signal beyond what rank-based softmax scoring already captures for models and problems of this scale. For this reason, and for clarity, we have not included the energy-scaled scores in tables.

4.2.4 Embedding-Based Conformal Inference

We apply embedding-based conformal inference methods to the 384-dimensional output of the SBERT model, before the fine-tuned classifier head, on which we used for softmax- and logit-based procedures. Our results show that the signal contained within the 384-dimensional output embeddings, taken directly from the SBERT model before the classifier head improves our conformal inference performance significantly for k -NN scoring, but has a significant negative effect when applying Centroid or Mahalanobis scoring measures.

Table 6: Embedding-based conformal inference results on AMuRD 384-dim embeddings (canonical split).

Method	Coverage	AvgSize	ClassCovGap	SSCV	SingletonHit	RawEmpty
Centroid (raw)	96.98%	11.509	2.80	36.67	58.33%	16
Centroid (cosine)	97.24%	11.968	2.85	37.42	57.58%	15
Mahalanobis (raw)	96.98%	11.524	2.80	36.67	58.33%	16
Mahalanobis (cosine)	96.85%	10.995	2.61	40.71	54.29%	12
k-NN (raw)	96.71%	1.081	3.53	4.68	97.28%	0
k-NN (cosine)	96.71%	1.081	3.53	4.68	97.28%	0

The prediction sets produced from Centroid and Mahalanobis distances-based scores span 11-12 out of 16 possible classes (Table 6) in order to reach the necessary marginal coverage guarantee. This is not an informative result, and is a key failure, especially when contrasted with the success of these non-conformity scores applied on the Fashion MNIST dataset (average set size 1.144 to 2.510, Table 2, out of 10 classes). These results show that a single summary statistic per class is a poor reflection of the representation space. The similarities between Centroid and Mahalanobis results indicate that the limiting factor in this procedure is not the lack of covariance correction, but rather a specific problem from reducing each class to a single representative point, regardless of the shape of the surrounding neighbourhood.

In contrast, k -NN produced the smallest average set size (1.081) of any non-conformity scores applied to AMuRD, and a low SSCV, demonstrating trustworthiness in the singleton sets produced. This is explained

by k -NN being non-parametric, and not reducing classes to single summary points so it is unaffected by the modality structure of a class in representation space and only assesses the nearby labelled points; a well-defined metric regardless of a class’s overall shape distribution.

We also find that raw-Euclidean and cosine distance produce identical results under the k -NN score to 3.d.p. for all assessed metrics. This is likely a consequence of the embeddings coming from the SBERT output, rather than a fine-tuned projection. Sentence-embedding models, such as `all-MiniLM-L6-v2`, are trained under cosine-similarity objectives so embedding magnitude is relatively uniform across examples and using signal gained from magnitude does not affect the results much (Table 6), and k -NN, which only looks at local structure, is completely unaffected.

We assess stability of performance over repeated trials (150 for Centroid/Mahalanobis, and 100 for k -NN), combining k -NN’s distance measures due to their equivalence, finding that the large average set sizes from centroid and Mahalanobis scoring are consistent throughout. We see that k -NN’s efficiency is stable, with a standard deviation of only 0.01 on its average set size of 1.08 (Table 7).

Table 7: Multi-trial summary, embedding-based scores (mean $\pm$ standard deviation across 150 trials for Centroid/Mahalanobis, 100 for k -NN).

Method	Coverage	AvgSize	ClassCovGap	SingletonHit
Centroid (raw)	96.2% $\pm$ 0.8pp	10.78 $\pm$ 0.76	2.51pp $\pm$ 0.45pp	60.7% $\pm$ 4.9%
Centroid (cosine)	96.2% $\pm$ 0.8pp	10.98 $\pm$ 0.84	2.88pp $\pm$ 0.45pp	57.8% $\pm$ 4.9%
Mahalanobis (raw)	96.2% $\pm$ 0.8pp	10.80 $\pm$ 0.76	2.50pp $\pm$ 0.44pp	60.7% $\pm$ 4.8%
Mahalanobis (cosine)	96.0% $\pm$ 0.8pp	10.02 $\pm$ 0.77	2.92pp $\pm$ 0.45pp	58.6% $\pm$ 4.9%
k -NN (both modes)	97.1% $\pm$ 0.4pp	1.08 $\pm$ 0.01	3.42pp $\pm$ 0.33pp	97.4% $\pm$ 0.3%

We suggest an explanation of these results by the semantic structure of the AMuRD dataset, reflected in the UMAP and t-SNE projections of the embedding space (6). AMuRD’s product categories can be described as internally heterogeneous, meaning that within a category such as "Sauces, Dressings & Condiments", the examples contained may span a wide range of dissimilar products, resulting in sentence embeddings which cluster into several distinct subregions, rather than one clear cluster. Despite the UMAP projection compressing a significant amount of information, we can hypothesise what products are contained in the sub-regions, based on their position; the embedding space makes semantic sense. Figure 10 provides an intuitive example of how true, genuine semantics do not naturally form clear clusters in embedding space, and demonstrates the calculated centroid sitting far from many class examples due to the wide spread semantics. This spread leads to an inflated conformal threshold, meaning examples far from the centroid still report that label, producing overly large prediction sets. This failure occurs on the raw SBERT representation, before applying the classifier head, demonstrating that how SBERT organises these semantics does not correlate with specific class clusters, and this is not a property introduced during fine-tuning.

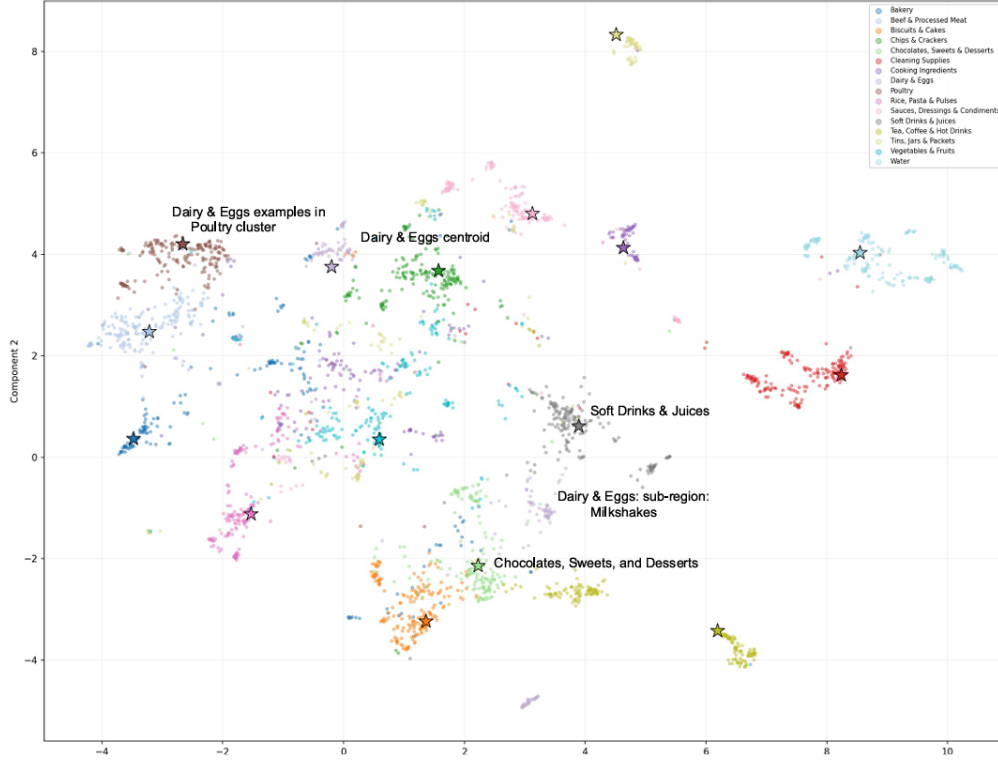


Figure 10: Demonstration of genuine within-class semantics of class “Dairy & Eggs” spreading across the embedding space through the AMuRD UMAP projection of the 384-dimensional embeddings, trained specifically for semantic meaning

We encourage future research in Section 6 to compare this pipeline (from 384-dim SBERT embeddings) with the same pipeline’s results applied to the 128-dim representation taken from the classifier head, to assess if task-specific fine-tuning affects class clustering in the embedding space, and consequently affects the embedding scores that reduce classes to a single summary statistic.

We briefly compare all the aforementioned methods through Table 8, and Figures 11, 12 to provide a clear comparison. These results motivate our choice of k -NN and LAC for clustering; k -NN demonstrates efficiency (small average set size) along with class-conditional unfairness via ClassCovGap. LAC’s multi-trial view shows a very large spread of coverage across trials, and the only one that has a significant portion of uncovered results.

The scoring methods we are interested in are LAC and k -NN, which we attempt to calibrate and improve performance for in Section 4.2.5.

Table 8: AMuRD single split master comparison table.

Method	Coverage	AvgSize	ClassCovGap	SSCV	WSC	SingletonHit	RawEmpty
APS	98.55%	1.845	3.45	4.19	90.79%	99.19%	22
Centroid (cosine)	97.24%	11.968	2.85	37.42	85.53%	57.58%	15
Centroid (raw)	96.98%	11.509	2.80	36.67	82.89%	58.33%	16
k-NN (cosine)	96.71%	1.081	3.53	4.68	86.84%	97.28%	0
k-NN (raw)	96.71%	1.081	3.53	4.68	86.84%	97.28%	0
LAC	97.24%	1.326	2.80	6.90	86.84%	99.13%	0
Mahalanobis (cosine)	96.85%	10.995	2.61	40.71	82.89%	54.29%	12
Mahalanobis (raw)	96.98%	11.524	2.80	36.67	82.89%	58.33%	16
RAPS	98.16%	1.749	3.38	9.29	89.47%	99.77%	14

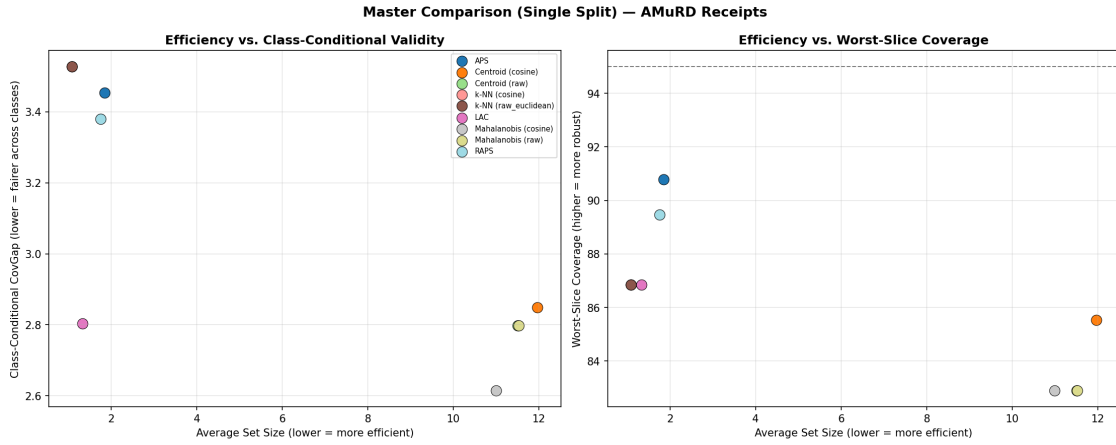


Figure 11: Master comparison of all nine scores tested on AMuRD plotting average prediction set size (lower is more efficient) against two independent measures of conditional coverage: class-conditional coverage gap (left; lower is fairer across classes) and worst-slice coverage (right; higher is more consistent coverage across random sub-populations) Centroid and Mahalanobis produce large sets with no compensation.

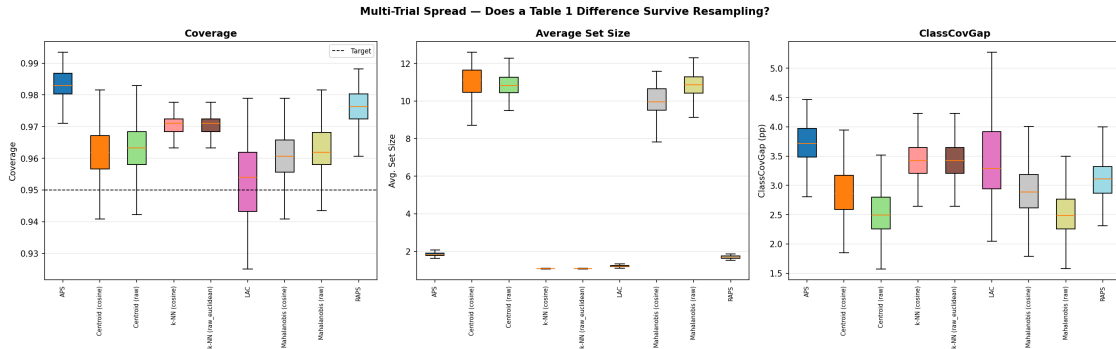


Figure 12: Coverage, Average set size, and ClassCovGap visualised across multiple trials (200 for LAC, APS, RAPS, 150 for Centroid, Mahalanobis, 100 for k -NN). Centroid and Mahalanobis exhibit large prediction sets. LAC is the only score to produce a significant amount of undercovered sets, and demonstrates the largest ClassCovGap spread. k -NN and APS both produce efficient, adaptive sets across all splits.

4.2.5 Classwise and Clustered Calibration on LAC and k -NN

From the results so far, we see that standard LAC undercovers Tins, Jars,& Packets at 92.31% and, despite the high efficiency, k -NN scoring produces the worst ClassCovGap across all the tested non-conformity scores, indicating the marginal coverage is hiding undercoverage on particular classes. We test whether classwise or clustered calibration (Section 2.8) improve these results, using a fixed clustering-fold quota of 20 examples per class and $M = 4$ clusters directly.

Table 9: Classwise and clustered calibration, LAC and k -NN, AMuRD (canonical split, $N_{cal} = 507$).

Score	Strategy	Cov.	AvgSize	ClassCovGap	Worst class	Worst cov.
LAC	Standard	97.24%	1.326	2.80	Tins, Jars & Packets	92.31%
LAC	Classwise	94.61%	1.653	3.86	Poultry	76.47%
LAC	Clustered	96.19%	1.528	3.72	Poultry	76.47%
k -NN	Standard	96.71%	1.081	3.53	Bakery	88.89%
k -NN	Classwise	96.98%	1.172	3.96	Poultry	88.24%
k -NN	Clustered	96.58%	3.832	4.43	Tea, Coffee & Hot Drinks	83.72%

There is clear instability of classwise calibration across both non-conformity scores. Both undercover the Poultry class, at 76.47% coverage for LAC (i.e. the true label is included in only 76.47% of prediction sets for Poultry inputs) and 88.24% for k -NN. Since LAC and k -NN are very structurally different (continuous softmax confidence vs. discrete embedding-neighbour agreement), this indicates that the issue stems from a property of the Poultry class itself. We find undercoverage of this scale surprising under classwise calibration due to construction. Since each class’s threshold is computed from an exchangeable sequence of just that class’s data, the rank argument from Section 2.4.2 should apply independently to every class, proving the marginal guarantee. However, this guarantee is an expectation over many calibration draws, and is particularly sensitive to a lack of calibration data. Under both LAC and k -NN, the threshold $\hat{q}$ for Poultry was computed under a draw of 31 examples. Since we choose $\hat{q}$ as the $\lceil (n+1)(1-\alpha) \rceil$ -th of the calibration set’s non-conformity scores, which under $\alpha = 0.05$ and $n = 31$, is the 31st position score, the threshold is just the largest non-conformity score available. Now this threshold is no longer a reflection of Poultry’s genuine difficulty as a class, but a random selection of the calibration example that happens to be hardest, which happens to sit at around the 76.47%-th quantile (LAC) of the test set data, and therefore produces prediction sets without the true label for 12 examples. The convergence under both LAC and k -NN suggests the difficulty distribution of Poultry examples has a large spread of difficulty, but it is the small sample size which prevents this from being averaged out, instead providing an unrepresentative threshold.

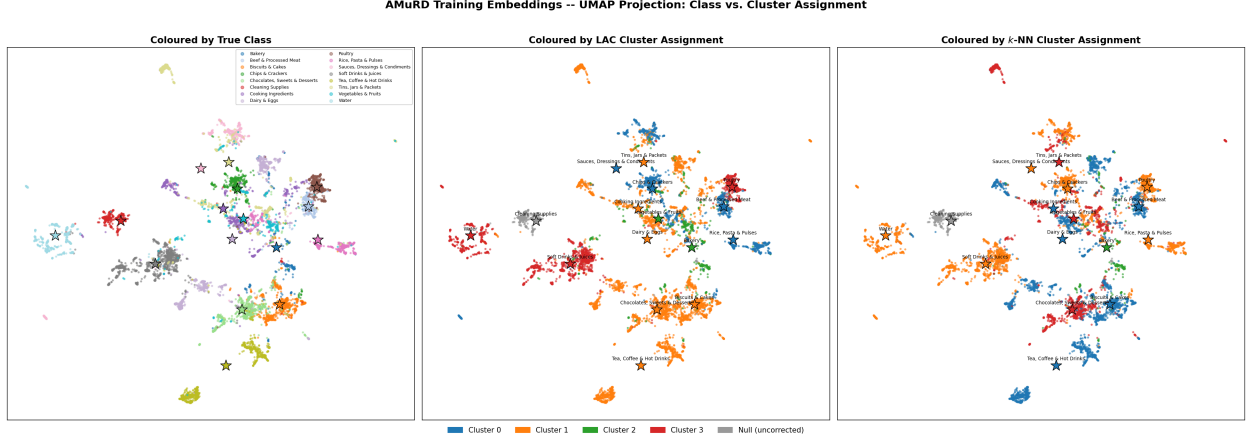


Figure 13: UMAP projection of AMuRD embeddings, coloured by true class (left), then coloured by the clusters each class was assigned by Ding et al.’s score-distribution clustering procedure on LAC (middle) and k -NN (right). Cleaning Supplies, Water, Soft Drinks & Juices and Tea, Coffee & Hot Drinks form visually distinct regions and the remaining 12 classes overlap. Water and Poultry are pooled together under both scores, although they seem different in the space, showing score-distribution similarity doesn’t imply geometric similarity

The clustering approach - in which classes are grouped into clusters with a shared $\hat{q}$ per cluster (Section 2.8) - has less consistent results across the scores. Under LAC, overall coverage returns to a valid level, and the class on which clustering was motivated, Tins, Jars, & Packets, is fixed. On the other hand, the Poultry class remains at the same level of undercoverage.

We explore the clustering mechanism intuitively through a UMAP projection in Figure 13. We have already identified that the UMAP does not demonstrate 16 clear-cut clusters; there are a few classes (Cleaning Supplies, Water, Soft Drinks & Juices, and Tea, Coffee, & Hot Drinks) where we can seemingly identify distinct class structure, but the rest forms an overlapping region with no visible boundaries between classes. This indicates that the representation space likely does not contain the separable geometric structure that clustered conformal inference assumes. LAC-based clustering groups together Water and Poultry, despite the embedding space cleanly isolating Water while Poultry sits in a crowded region of other classes. Pooling these two classes reflects removing the differentiation between aleatoric uncertainty (Poultry) and epistemic uncertainty (Water), and a single threshold applied to these two will not improve the results. The clustering from k -NN creates a 6-class cluster (Cluster 0), in which the classes are geometrically dissimilar, in turn producing a pooled threshold which reproduces the global threshold, and changing coverage for none of its members. This indicates that the cluster is too large to accurately represent any true sub-structure at all.

This UMAP-based evaluation is not necessarily conclusive, since the reduction in dimensionality to produce the UMAP can both create and remove apparent structure, so distances between well-separated regions are not meaningful. The figure does demonstrate that AMuRD’s classes are not all separable, and a cluster-

ing mechanism based on score distribution alone has no way to distinguish between classes which are equally difficult, but for entirely different reasons, as seen with Water pooled with Poultry.

We find here that this clustering approach is therefore not a reliable fix for AMuRD’s poor conditional coverage. Any improvements are dependent on the score it is applied to, and whether the classes it happens to group are similarly difficult, or too different to share a threshold. We consider the idea that clustering directly on the embeddings may improve our results.

4.2.6 Clustering Embeddings Directly: An Exploration

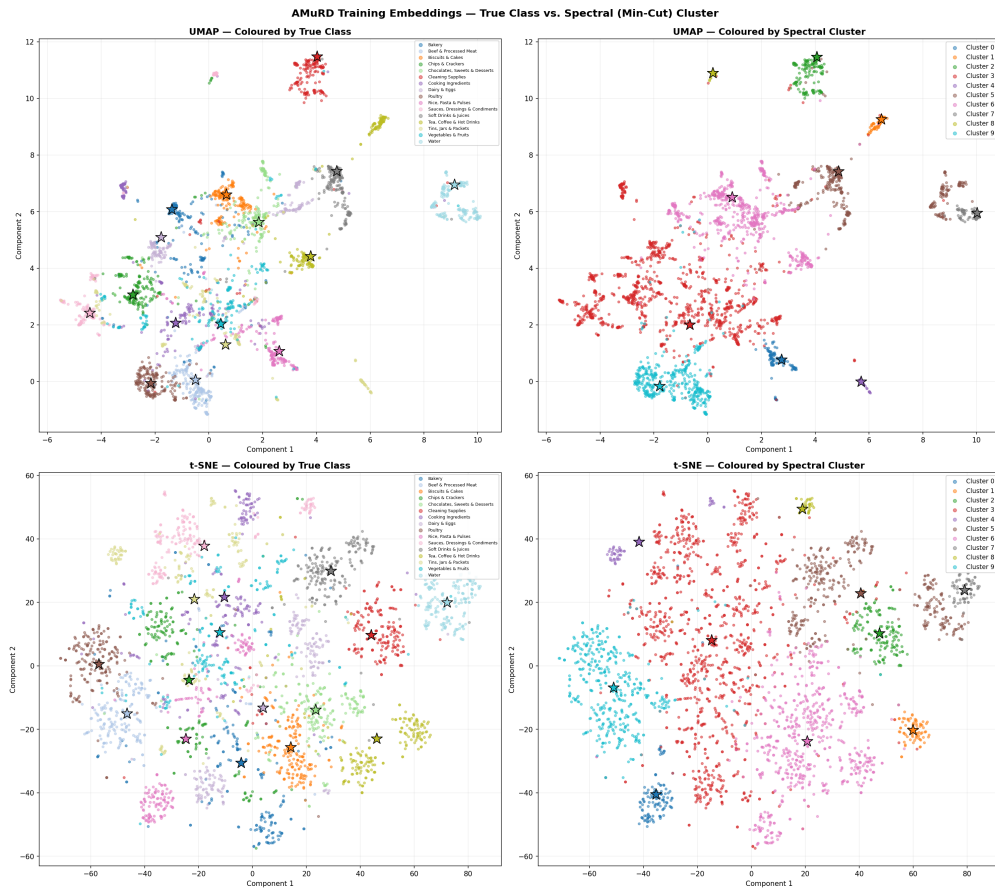


Figure 14: AMuRD training embeddings, coloured by true class (left) and by min-cut determined clusters (right), visualised via UMAP (top) and t-SNE (bottom) projections.

The previously explored clustering methods, following Ding et al., group entire classes together based on the similarity of their non-conformity score distributions. This process means there is no way to treat two points from the same class differently, even when their position in representation space indicates very different levels of uncertainty; a key limitation given the UMAP evidence above (Figure 6a) demonstrates that AMuRD is not sixteen separable class-shaped regions, but rather a few isolated categories alongside a large

blend containing most of the remaining classes. This motivated us to explore the possibilities of clustering the embedding points themselves, rather than the classes, and calibrate a threshold per geometric region instead of per class. A comprehensive exploration of this idea is beyond the scope of this thesis. It raises many questions; which clustering algorithm to use, how to assess similarity, how many regions of embedding space to cluster, how to preserve the class-level information once a region spans several classes, and how to rigorously validate a conformal threshold across scores and datasets. We undertake an exploratory test of this intuition, rather than a finished method, demonstrating a “first-step” towards what would be required of a full treatment, detailed in Section 6.

We apply a min-cut clustering algorithm to our embedding space. A min-cut partition divides the points into groups by finding the sparsest possible boundary between well-connected regions. This is done by treating the entire embedding space as a graph where each point is a node connected to its 15 nearest neighbours, and edges are weighted by similarity to their nearby points. Min-cut clustering then finds the way to separate this graph into 10 groups, while breaking the least total connection strength, essentially finding the sparsest boundaries to separate the space. Since exactly solving this is intractable at scale, we approximate it via spectral clustering: decomposing the graph’s Laplacian and clustering the resulting eigenvectors, the standard, tractable relaxation of the min-cut objective. Since this method is independent of true class labels, we theorised it would separate a genuinely isolated class from a tangled group without knowing the true boundaries. A common failure of min-cut, especially when partitioning into many groups, occurs when it isolates a single, weakly connected point in sparse embedding space, minimising the strength of boundaries broken but producing a useless cluster. We use Normalised Cut [36] which rewards substantial clusters and discourages small clusters, therefore creating a meaningful partition into 10 clusters.

Figure14 compares the min-cut clusters against the true class groups. We find that one region groups most of the tangle of classes, a demonstration of the algorithm not forcing boundaries where none exist, whereas the well-separated groups form separate clusters, recovering real structure through the clusters.

Table 10: Standard versus region-clustered (spectral min-cut) calibration, k -NN score, AMuRD.

Strategy	Coverage	AvgSize	ClassCovGap	WSC	RawEmpty
Standard	96.71%	1.081	3.53	86.84%	0
RegionClustered	97.11%	1.126	3.17	88.16%	8

The observations from the UMAP and t-SNE projections are not completely reliable, due to the flattening which occurs to form the projections. We test a conformal pipeline on these clusters, calculating a k -NN based conformal threshold on each of these clusters, specifically assessing whether this method improves worse-case

coverage, measured through worst-slice coverage (WSC). This metric was chosen as it evaluates coverage through random projections of the input space, providing an independent check on whether our calibration method improves performance instead of using a metric which is directly correlated to the underlying regions used to build the partition. In Table 10 we observe a small improvement in WSC in comparison to standard conformal inference, but the results are fairly inconclusive, with this improvement coming with 8 empty sets, and a slightly larger average set size.

This brief experiment purely serves to answer whether this research direction - partitioning AMuRD’s embedding space by geometry rather than class - is worth developing further. The small structure recovered is genuine and interpretable, and we see a slight improvement on the worst-case metric, but a full comprehensive exploration (systematic selection of the number of regions, handling of regions spanning across classes, validating the approach across non-conformity score and datasets) is left for future work (Section 6).

4.2.7 AMuRD Summary

Our experiments on AMuRD first aimed to explore whether the pre-defined conformal pipeline continued to produce usable prediction sets. We found that the LAC and k -NN scoring methods produced efficient prediction sets at valid coverage, with this result holding stably across many repeated calibration draws. Our second research question explored whether the embedding space contained exploitable signal we could use to improve the quality of our prediction sets. This result depended entirely on the process of signal extraction. The fully non-parametric method of k -nearest-neighbours produced the best performance across all non-conformity scores, whereas the global summary methods, Centroid and Mahalanobis, offer little improvement. We investigated standard calibration methods, aiming to improve conditional coverage for two non-conformity scores, finding that a lack of calibration data in classes - Poultry specifically - resulted in undercoverage. Our chosen clustering approach based on class’s score distributions was also unsuccessful, a result attributed to the lack of natural clustering in the classes in the embedding space. A brief exploration of clustering on embeddings provided a small improvement, indicating this as a future research direction.

These results reflect the difference in the internal structure of AMuRD compared to Fashion MNIST. The multi-modal class structures in AMuRD must be treated in a different way, especially as this representation space is trained purely on semantic similarity, rather than with an expressed classification goal in mind as in the previously applied CNN.

4.3 Invoices: The Target Dataset

We continue the exploration of our two recurring questions of this chapter, whether we can achieve usable prediction sets with conformal inference, and whether the embedding space contains signal beyond the detail contained in the softmax output, now applied to our target dataset, real data rather than a research-ready dataset.

4.3.1 Dataset and Model

Introduced in Section 3.1, the dataset explored here is a confidential dataset consisting of 2,169 invoices across 12 accounting categories, embedded using the frozen `all-MiniLM-L6-v2` SBERT model to 384-dimensional representations. Class sizes are highly imbalanced, ranging from 705 invoices (Purchases) to 45 (Stationery & Postage). One category in particular, Accountancy Fee, warrants specific mention: 91% of its invoices originate from a single vendor, since almost every client represented in this dataset uses the same accountancy firm for their own bookkeeping. Geometric analysis of the embeddings [18] demonstrates these embeddings forming a very tight cluster. In the following results, we see this affects the behaviour of this category.

We apply a fine-tuned classifier head on top of the frozen SBERT embeddings. Under training on the stratified split (1,298 reference/ 431 calibration/ 440 test), we reach 93.18% test accuracy. We’ve used a non-linear classification head with two linear layers and ReLU and dropout.

4.3.2 Softmax-Based Conformal Inference

Table 11 shows the behaviour of softmax based conformal inference across 200 independent calibration/test splits. The marginal coverage behaves as expected in the literature [6], with LAC’s marginal coverage close to the target (95.71%), and APS and RAPS both overcovering as a result of the discreteness in the data combined with limited calibration data (Section 3.5).

Table 11: Softmax-based conformal inference on the invoice dataset, 200-trial means. Target coverage 95%.

Method	Coverage	Avg. Size	ClassCovGap	FracUnderCov	SSCV
LAC	95.71%	1.110	40.7pp	37.7%	6.0pp
APS	98.72%	2.316	14.0pp	18.5%	−1.1pp
RAPS	98.16%	1.786	15.0pp	25.8%	2.3pp

LAC’s average ClassCovGap (the worst-case difference between a class’s observed coverage and the target coverage) is 40.7 percentage points, and reaches a maximum of 81.7 pp across the 200 trials. This is a consequence of small sample size, meaning that coverage is restricted to a discrete set of values. If we have n examples in the test set, coverage can only take values $0/n, 1/n, \dots, n/n$, so each additional covered example

results in another discrete jump on coverage. If n is large, such as with Purchases with 705 invoices, coverage only increases by $\approx 0.17\text{pp}$ per covered example. Therefore, high ClassCovGap is a direct result of the class imbalance rather than score failure.

We also see a negative result for APS’s SSCV. SSCV (Size-stratified coverage violation) measures the largest deviation from 95% across partitions by prediction set size (Section 3.5), and a negative result here means that APS’s worst covering partition still exceeds 95%. This is consistent with APS’s overcoverage, in which the non-conformity score produces conservative prediction sets both on average and across all set sizes. This result doesn’t come without cost; APS produces average sets of size 2.316, the largest of the softmax based scores, showing that the safety margin which guarantees coverage comes at the cost of efficiency.

We also see large set sizes from RAPS, where 29.8% of test inputs received a set of size 3; a result of the $k_{reg} = 3$ regularisation boundary where the penalty term starts to apply. Implementing this penalty has forced prediction sets to size 3 and removed adaptivity: the size 3 sets are not a true reflection of the underlying uncertainty.

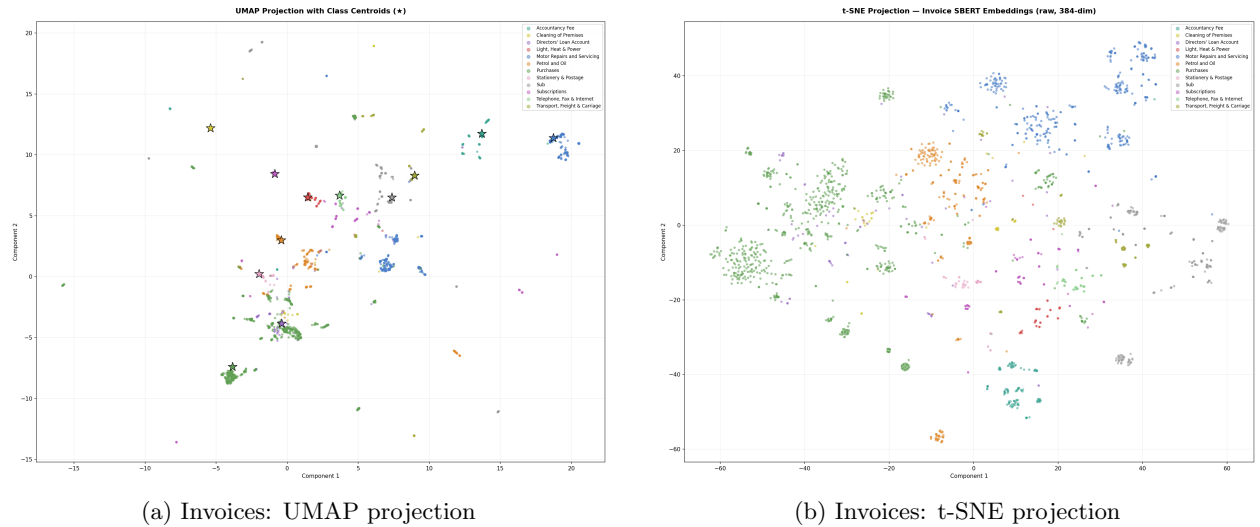


Figure 15: 2D projections of the invoice data’s 384-dim training embeddings taken from the SBERT output, coloured by class, comparing UMAP with t-SNE as two dimensionality-reduction techniques. The consistent structure indicates that the observed class relationships are genuine, and not a result of either projection method. Mean class embeddings (Centroids) are marked with stars in UMAP. Note that distances between clusters are not meaningful in t-SNE.

4.3.3 Logit-Based Conformal Inference: Energy Scaling

In applying energy scaling here, we observe results consistent with the two previous datasets, where the auto-tuning process of β selects the smallest available value, resulting in energy-scaled variants of APS and

RAPS which are statistically indistinguishable from the base scores (APS: 98.86% coverage / 2.282 average size, APS+Energy: 98.86% coverage / 2.286 average size, RAPS: 98.86% coverage / 1.807, RAPS+Energy 98.86% / 1.814). This finding has now been replicated across all our datasets, and through both image and text data, which is evidence to suggest that there is genuinely little exploitable signal in the logit magnitudes beyond what is captured by APS and RAPS from the ranked softmax scores.

4.3.4 Embedding-Based Conformal Inference

Now we explore the effects of including signal taken from the embedding layer of the model pipeline. As seen in Table 12, we obtain results mirroring that of AMuRD’s, in which k -NN scoring under either Euclidean or cosine distance provides the strongest results out of any competing non-conformity score from the rest of the chapter. k -NN produces prediction sets of average size 1.189 with coverage level 98.86%, both efficient and valid. This is another demonstration that k -NN’s fully non-parametric construction outperforms scores that produce global summaries of class shape, especially on unevenly distributed reference sets.

Table 12: Embedding-based conformal inference, invoice dataset. 12 possible classes.

Method	Coverage	Avg. Size	ClassCovGap	SingletonHit
Centroid (Euclidean)	97.27%	4.991	5.0pp	81.3%
Centroid (cosine)	97.27%	5.039	5.0pp	82.4%
Mahalanobis	100.00%	10.385	5.0pp	100.0%
k -NN (Euclidean/cosine)	98.86%	1.189	8.3pp	99.7%
k -NN (Mahalanobis)	97.50%	1.130	28.3pp	97.9%

Centroid and Mahalanobis distance once again produce large prediction sets (average ≈ 5 and ≈ 10 respectively) a result we can intuitively understand from looking at the UMAP projection; centroids are not neatly in the centres of class clusters, but rather scattered, and often between or far from sub-regions of their class. This inflates scores even for examples which may be a typical representation of that label.

On this dataset, as a further investigation due to the large class-imbalance, we also apply the Mahalanobis metric (Section 2.7.2) (scaling the space by its covariance before reading distance) to the k -nearest-neighbours non-conformity score. We find the results are significantly worse than k -NN with either Euclidean or cosine distance measures, particularly looking at ClassCovGap (28.3pp in Table 12). The covariance introduced when we use Mahalanobis is subject to the reference set which we have established to be messy; the imperfectly-conditioned metrics can distort which points register as ‘nearest’ to the example in question in a way not-uniform across classes.

Given k -NN’s performance despite the lack of clarity in the embedding space, we explore these results

further, to test if they truly reflect good performance.

We have $k = 5$ in our implementation, and 93.4% of test invoices have at least one invoice in the nearest five from the same vendor, with an average of 81.1% (four out of five) of an example’s neighbours share its vendor. This observation reflects how the representation space is structured; the nearest neighbours of a particular point are those from the same vendor as that point. This makes intuitive sense; the semantic similarity should be correlated with the company, however this does have an effect on how the score understands the representation space. Since the embedding space is encoding many near-identical invoices, the k -NN score can assess a new test example by considering if there are similar examples already in the reference set, rather than building a true conceptual understanding of the example’s semantics. Therefore, k -NN is essentially recognising recurrent vendors, and producing very efficient prediction sets based on this knowledge base. This does not align with the geometric motivation we described in Section 2.7.2 in which k -NN was introduced to avoid making assumptions about the global structure of the embedding space, rather than a memorisation procedure.

4.3.5 Data Limitations

This dataset comprises of 2,169 invoices, which is then split further into the train, calibration, and test splits, a substantial difference from AMuRD (10,000+) or Fashion MNIST (60,000+). The lack of data points has a specific effect on the conformal inference procedure; the counts in calibration and test sets for each class are small enough that per-class coverage can only take a few discrete values, which results in unstable per-class coverage estimates which are a property of the data rather than the mechanism we are trying to assess. We also have a severe class imbalance (45 to 705 across classes). Despite AMuRD also being imbalanced across classes, the invoices have the added challenge of the smallest classes not having enough examples to provide reliable quantile estimates.

Aside from the issue of data volume, we have another limitation in the concentration of vendors and the large amount of near-identical examples. This affects our lack of data even more, as a notable fraction of the 2,169 examples are near-repeats, rather than independent examples of what different categories look like, distorting what the conformal inference calibration set actually represents. This poses the question; could we achieve strong accuracy on the label from only recognising the vendor, rather than having any genuine understanding? This is explored in [18]; a vendor-only baseline that maps each vendor to its most frequently observed category achieves 89% accuracy. This confirms that vendor recognition is a strong predictor of label, independent of any genuine semantic understanding.

Another limitation in this dataset is the aleatoric uncertainty, semantic ambiguity that cannot be recovered from the invoice alone. The overlap between Purchases and Director’s Loan Account (seen in Figure

15a) is due to the distinction between the two categories simply being who paid in a specific case (Director or business) [18]. No amount of model training or additional data can improve this ambiguity as it is intrinsic to the task. A good conformal inference procedure would provide prediction sets of size two ($\{\text{Purchases, Director's Loan Account}\}$) for any which fit into this ambiguity. Due to these limitations, we avoid clustering methods here as they would only bring more instability due to using even smaller groups in the data.

4.3.6 Invoices Summary

We have once again demonstrated that conformal inference can produce efficient and usable prediction sets; the softmax-based outputs calculated from the classifier head all delivered coverage at or above target with efficient average set size. Following on to our second research question, the embedding space has proven to contain usable signal, especially in the case of k -NN's performance.

There are specific limitations we have uncovered. LAC's class-conditional stability is highly dependent on class size, and the highly concentrated Accountancy Fee class provides a large amount of instability in every class-conditional metric through this dataset. RAPS is limited here by the default parameters, with an inflated number of prediction sets of size 3 as a result.

5 Discussion

5.1 Summary of Findings

The goal of this research was to test whether conformal inference could replace the overconfident outputs of transformer based classifiers by producing calibrated prediction sets carrying formal coverage guarantees, and whether these outputs could be trusted on a per-example basis. We investigated eleven non-conformity scores across three datasets, as well as classwise and clustered calibration methods, to determine whether conditional coverage was achievable as well as marginal coverage. Marginal coverage was reliably achievable with an exchangeable calibration/test split, so checking validity functioned as a sanity check, and a way to add depth of information to the other metrics reported. We initially hypothesised that the earlier model layers - logits and embeddings - which contain more information about the underlying input would in turn produce more accurate prediction sets. This was not the case with logit-based conformal inference; across all experiments the logit-based energy scaling offered no improvement to the prediction sets quality. The improvements from embedding-based measures came from the non-parametric score k -NN, a result of class shape in representation space proving too internally heterogeneous for a single number summary. However, the performance of k -NN on the invoices dataset appeared to be a result of the procedure recognising

near-duplicate invoices from the same vendors, a weakness that the parametric scores were not as open to. Calibration approaches improved conditional coverage only if the score contained the necessary resolution.

We summarise our results by their contribution to answering our core research questions laid out in the Introduction.

5.1.1 RQ1: Can conformal inference provide reliable, calibrated uncertainty estimates for overconfident neural network classifiers?

The conformal inference procedure is consistently reliable at the marginal level ($\sim 95\%$ of prediction sets will contain the true label) but this result does not imply that any particular class, or any individual prediction set is reliable. We find that reliability does not hold to the class conditional level; there is likely to be a difficult class in which the prediction sets are less likely to contain the true label, and we explore methods to improve this result RQ3. We have aimed to go beyond observing the failure in conditional reliability, and explain why it fails.

Since any method can trivially produce reliable and valid sets by consistently including the vast majority of classes, we go further to ensure our prediction sets are adaptive, responding to the genuine level of uncertainty of a specific input. Our results show that the most simple and efficient scores tested were consistently the least adaptive; producing small sets on average, but without those sets growing with increased uncertainty. Scores that were less efficient (larger average set size) proved to track difficulty meaningfully, demonstrating that the average set size metric hides nuance. Overall, conformal inference reliably delivers valid prediction sets, but well-calibrated prediction sets are a property of the chosen score function, and are not consistent across different datasets. We address the question of reliability holding at the class level in Section 5.1.3 .

5.1.2 RQ2: Does the internal geometry of a classifier’s embeddings explain and predict classification difficulty?

Embedding geometry does carry genuine, usable information about classification difficulty, but when accessed in a specific way. Firstly, ensuring the distance measure being applied is exposing the relevant structure in the geometry; for example where difficulty of embeddings is represented through magnitude, cosine distance will discard this information. A more important result lies in the type of non-conformity score we use to perform conformal inference. We find that reducing classes to single summary statistics is unreliable when working with real, likely internally heterogeneous, data, whereas purely focusing on the local neighbourhood around a test point can uncover signal in the geometry, and produce significantly better conformal results, especially in a more complex embedding space. Looking closely at the real data used, we note that geometric

proximity between points can encode an administrative pattern in the same way it encodes genuine, task-relevant semantic difficulty. Therefore, geometric results are evidence that the geometry is predictive, but not necessarily evidence of genuine input understanding.

5.1.3 RQ3: Can specific clustering methods applied to the embedding space improve conformal inference performance?

Our results show that clustering can substantially improve conditional coverage, but only in certain circumstances. With per-class calibration set size below a certain threshold, our conformal threshold is no longer an estimated quantile but fixed at whichever calibration example happens to be the most difficult, conforming the least to the class under our chosen measure. This fragility is easily triggered by imbalanced, realistic datasets, and the clustering method we have demonstrated aims to circumvent this issue by pooling calibration data across classes. We find that this method is not always applicable; the underlying data may not possess any meaningful clustering structure that the algorithm can exploit. Clusters can form between classes which are “difficult” for different reasons, or can form where there is no true structure. The benefit from clustering has to be extensively tested for a given dataset and score, and cannot be assumed, giving us a slightly inconclusive answer for this question. We indicate future directions of research in Section 6.

5.1.4 RQ4: Which properties of a particular method indicate genuine practical value to a user relying on its predictions?

The properties which indicate genuine practical value of prediction sets for a user are adaptiveness and honesty. Adaptiveness ensures that prediction set size is a tracker for genuine difficulty. Variable set size will provide a user with an obvious signal to rely on, in which larger prediction sets are reviewed more extensively, without them having to track a probability. We measure it through the adaptiveness gap and through SSCV (does the coverage guarantee hold across set sizes). Methods may be adaptive in only one of these senses; RAPS on AMuRD produces variable set sizes but with high SSCV, meaning these set sizes are not a reliable indicator of whether the true label is present.

With “honesty”, we aim to produce trustworthy singleton sets, and for the model to prioritise signalling genuine uncertainty rather than producing a plausible single answer which is incorrect. For a user, if this is satisfied and we are confident in the quality of the singleton sets, these can progress down an automated pipeline, where flagged prediction sets are set aside for human intervention. Honesty ensures that any model errors are visible. This is also why we report the empty-set-rate (and argue against the force-inclusion fix in practice) as singleton sets which are truly empty sets are not honest.

Table 13: An informal practical-value assessment of each non-conformity score.

Score	Adaptiveness	Honesty	User Summary
LAC	Low	High	Small, trustworthy sets, but set size doesn't correlate to difficulty
APS	High	Average	Produces adaptive sets with uniform coverage at the cost of efficiency (large average sets)
RAPS	Average	Average	Adaptive on Fashion MNIST, unreliable on AMuRD; sensitive to untuned hyper-parameters
Centroid	Low	Low	Large sets, not from input difficulty
Mahalanobis	Low	Low	Large sets, not from input difficulty
<i>k</i>-NN	Average	High	Highest efficiency and singleton reliability of any score tested, but sensitive to non-semantic local structure - i.e. invoices

A better assessment of practicality would be derived from observed user behaviour, these results are only from our assumed correlation between practicality and the behaviour of the sets.

5.2 Limitations

Data Volume The majority of deep-learning research uses large datasets, and this is where our process is particularly limited. AMuRD and the invoice dataset are both too small to produce stable calibration thresholds, as we see particularly in Section 4.2.5, applying classwise calibration for AMuRD. This could have been remedied through a re-shuffle of the training/calibration/test split, which we did not change from the original split, with $\sim 90\%$ of data in the training set. For future work it is important that, for well-performing conformal inference, there are enough calibration examples (≥ 39 for $\alpha = 0.05$) in order to set $\hat{q}$ as a genuine quantile instead of a chance highest value.

Methodological Shortcuts We have made several suboptimal methodological choices for the sake of convenience and efficiency, which are limitations on the results. We avoided hyperparameter tuning for RAPS by selecting the default $k_{reg} = 3$, $\lambda = 0.2$, which likely resulted in the high SSCV observed on AMuRD. The clustering configuration set up in Section 4.2's classwise/clustered analysis was chosen manually as a cheap fix to avoid using the standard formula which implicitly assumed a significantly larger dataset. A better approach would be a higher-level tuning process, sweeping across cluster counts and quota size. Also, the quantile resolution we used to determine the classes similar enough to pool was equal to the number of raw calibration scores, meaning there was no summarising of score distribution. Therefore, two classes with genuinely similar difficulty could look dissimilar, and avoid being pooled together, purely because of the examples in the calibration set. This could be further explored by dropping our quantile resolution.

Post-Hoc Application More limitations to providing a complete assessment of our research questions come in the scoping choices we have made prior to conducting research. All our conformal procedures were applied post-hoc to a single classifier model per dataset (Section 3.2), and do not consider that results may be improved from a different model, or from a deployment which continually retrains the classifier to optimise classification and uncertainty quantification. We chose to implement split conformal inference (Section 2.4) over its full or adaptive variants, therefore assuming a static and exchangeable population. This means our results will not generalise to a live deployment in which we may have a continuously arriving stream of invoices, many of which may be unfamiliar, without further work.

6 Future Work

6.1 Clustering

The key direction we encourage for future research is to develop the exploratory experiment of Section 4.2.6, clustering embeddings directly rather than classes.

The clustering procedure we have implemented groups classes based on the similarity of their non-conformity score distributions but, as seen in the results on AMuRD, this is not necessarily the right level on which to partition the inputs. From the UMAP projection, we see that the AMuRD data does not form 16 well-separated clusters, but instead see that the majority of examples across most classes form a tangled region. Class-level partitioning cannot work with this spread, as it has no way to treat examples belonging to the same class differently, even if some form a tight cluster in an isolated region of embedding space, and the others are mixed into the tangle. These cases likely need a different conformal threshold. We propose partitioning the reference embedding space itself instead of the classes within it, calibrating a threshold per region rather than class. The intuition behind this idea came from wanting to identify and separate out the “messy” section of embeddings, giving it its own threshold, so the ambiguous points stop inflating the prediction sets of clear examples. Our min-cut experiment (Section 4.2.6) shows this could be feasible, and offers a small improvement in worst-slice coverage.

For future work, we identify four questions that must be resolved;

1. Which embedding clustering algorithm to apply, and using which non-conformity score? This selection requires a prior understanding of the representation’s geometry.
2. How many clustering regions to specify? We chose an arbitrary number of clusters, but a principled selection is needed, and cannot use the calibration scores it will be validated against during its computation.

3. How can we preserve class-level information when a region spans over several classes? We want to provide both regional-level and class-level conditional coverage.
4. How can we validate our process, ensuring no data leakage? Geometric partitions to assess conditional coverage may be correlated with the partition itself. We use WSC to introduce randomness, but a full validation would require results over multiple scores and datasets.

By fitting the partition on the training set rather than a fold of the calibration data, we preserve validity while avoiding further splitting of our data.

6.2 Does Task-Specific Fine-Tuning Reshape the Manifold?

Our embedding-based results on AMuRD are computed from the raw, 384-dimensional, frozen SBERT output, encoding semantic similarity rather than task-specific class separation. We have not looked at the 128-dimensional representations produced by the trainable classifier head. A direct comparison of the conformal inference pipeline applied to these two representation spaces would provide us with more nuance to our RQ2 answer. If task-specific fine tuning reshapes the manifold, the classes should form more unimodal shapes than in raw SBERT form, which should mean that centroid-based non-conformity scores could potentially be more powerful. A UMAP and t-SNE projection of AMuRD’s 128-dimensional embeddings from the classifier head (Appendix 1) demonstrate these uni-modal clusters, and gives strength to this hypothesis.

6.3 Towards Deployment

A concern during the work on the invoices dataset was the repeated, near-identical documents from the same vendors across the reference and test split. This could mean the performance of our conformal pipeline is a result of vendor recognition rather than a reflection of genuine model understanding. A potential data split based on vendor name would remove any leakage across calibration and test, proving how much of the performance is from genuine model understanding and not from memorisation, but would result in a test set of OOD examples. We note that all results in this thesis assume a single exchangeable population, something a genuine deployment in which invoices arrive continuously would violate. Testing on OOD examples would be a more genuine reflection of deployment, in which invoices will diversify over time. To ensure our results remain solid for this scenario, and uncertainty is reflected for new, unseen invoices, we encourage research looking into applying adaptive conformal inference, which updates the effective target coverage level by observing recent coverage.

7 Conclusion

The opening motivation in this thesis observed that the growing automation of administrative tasks must be weighed against the cost of the errors that automation introduces, and that this weighting depends on the trust we can place in a system’s confidence. Conformal inference, examined here across three datasets, eleven different non-conformity scoring methods, and two conditional-coverage adaptations, approaches the level of trust we look for, albeit partially. The marginal guarantee promised is easily achieved, but the conditional, per-class reliability that is key in practice (i.e. for an accountant reviewing one flagged invoice) is not consistently achieved. This thesis demonstrates several mechanisms where reliability fails while the overall coverage looks good; small-sample threshold failure, information lost under distance measures, dependence on cluster-compositions, and a specific genuine administrative confound sitting inside an apparently strong result.

We believe there is promise in the conformal inference landscape, the provision of a rigorous guarantee is not overlooked, but a significant amount of research is encouraged. A standardised procedure for determining which conformal method will support; every method we have assessed is sensitive to the underlying conditions.

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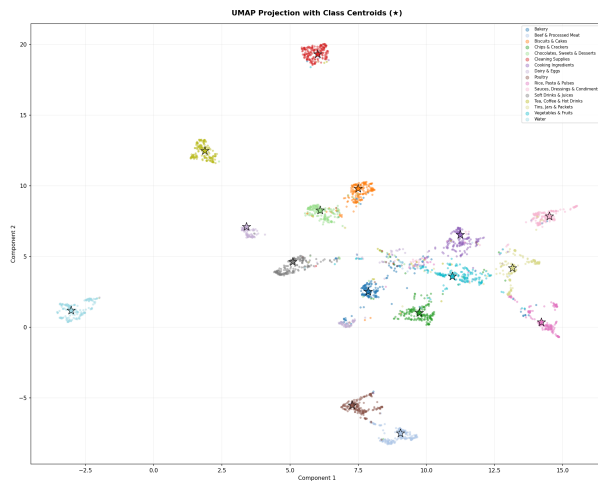
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8 Additional Figures



(a) 128-dim UMAP projection



(b) 128-dim t-SNE projection

Figure 16: 2D projections of the AMuRD 128-dim training embeddings taken from the classifier head. Demonstrates the uni-modal structure provided through a fine-tuned classifier head. Mean class embeddings (Centroids) are marked with stars in UMAP. Note that distances between clusters are not meaningful in t-SNE.